PolyField Track

System Operations Manual

Complete guide to installation, setup and operation

Version 4.0.3

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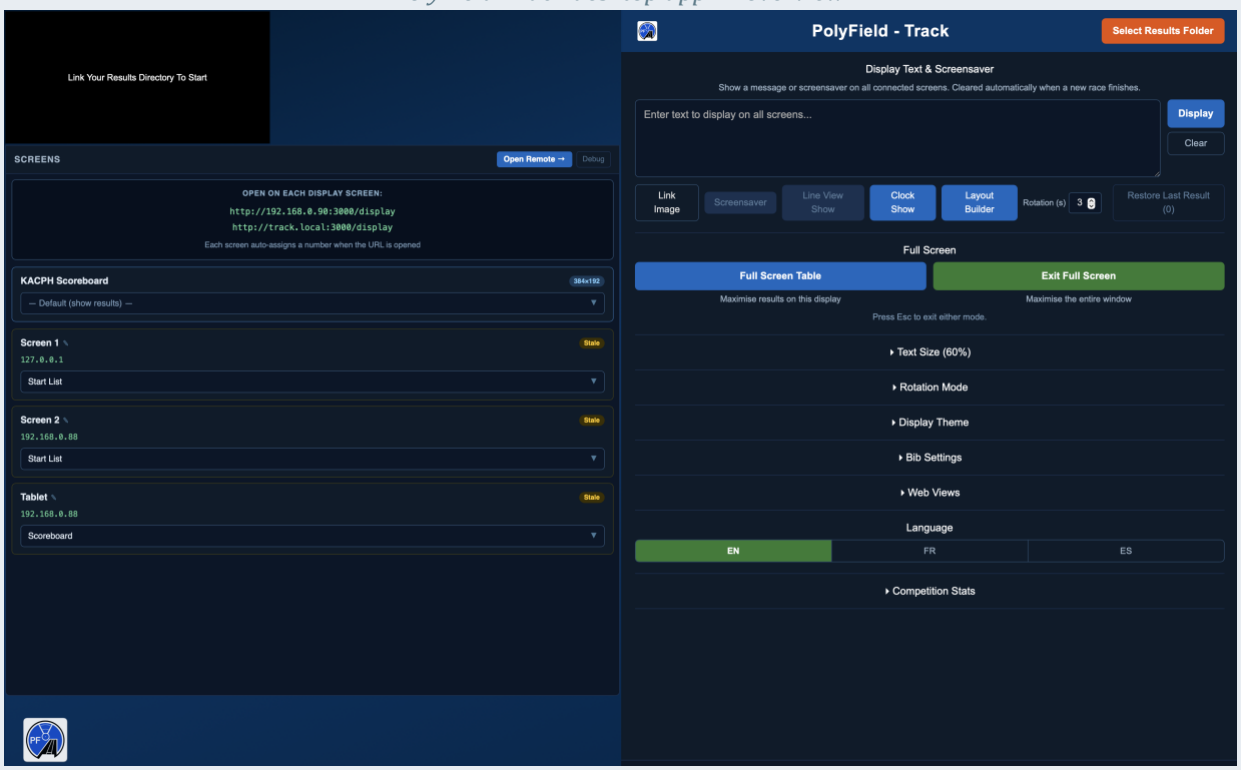
1 Introduction

PolyField Track is a real-time scoreboard and display management system designed for athletics meetings. It connects directly to FinishLynx timing equipment and distributes live results, timing data, and operator-controlled graphics to any number of display screens around the venue — instantly and automatically.

The system requires no configuration on display devices. Any browser-capable device on the local network can become a display screen simply by opening a URL.

PolyField Track desktop app — overview



1.1 What PolyField Track Does

Feature	Description
Receives results	Monitors a directory for new FinishLynx .lif result files and distributes them to all screens within seconds of a race being saved.
Live timing	Receives the FinishLynx running clock, event name, and wind data via UDP and drives real-time clock and event name displays.
Multi-screen	Manages any number of display screens independently. Each screen can show a different layout simultaneously.
Custom layouts	A drag-and-drop Layout Builder lets you design and save layouts combining results, clocks, logos, and other widgets.
Operator overlays	One-click text messages, screensaver images, and photo finish views can be pushed to all screens instantly.
Web access	All display pages are accessible from any browser on the network — no

	app installation required on display devices.
Analytics	Athlete search, speed dashboard, competition stats, and RAZA para-athletics rankings are built in.
Social graphics	Generate branded competition statistics and speed comparison graphics, exportable as PNG.

1.2 System Requirements

Requirement	Details
macOS	12 Monterey or later (Apple Silicon and Intel, universal binary)
Windows	Windows 10 or later (64-bit)
RAM	4 GB minimum, 8 GB recommended
Storage	200 MB for application; results directory size depends on meeting
Display	Any resolution; app window minimum 900 × 600 px recommended

Note: PolyField Track runs exclusively on the timing operator's computer or a dedicated screen control computer. Display screens (TVs, tablets, laptops) only require a web browser.

1.3 Network Requirements

PolyField Track creates a local HTTP server on port 3000. Display screens connect to this server over the local network.

Item	Notes
Local network	All devices must be on the same Wi-Fi or wired LAN
Port 3000	Must be open on the timing computer's firewall
mDNS / Bonjour	Enables track.local hostname — supported on macOS/iOS/most Android. Windows may need IP address.
Internet access	Not required for operation. Only required for Ollama WPA classification (optional) and initial licence verification.

Tip: If track.local does not resolve on Windows display devices, use the server's IP address instead (shown in the Screens panel).

2 Installation & First-Time Setup

2.1 Installing the Application

macOS

1. Download PolyField-Track-mac.zip from the GitHub releases page.
2. Double-click the zip to extract PolyField-Track.app.
3. Drag PolyField-Track.app to your Applications folder.
4. On first launch, right-click the app and choose Open to bypass Gatekeeper, then click Open in the dialog.

Warning: macOS may show "unidentified developer" on first launch. This is expected for unsigned apps. Right-click → Open is only required once.

Windows

5. Download PolyField-Track-windows.zip from the GitHub releases page.
6. Extract the zip and run PolyField-Track.exe.
7. If Windows Defender SmartScreen appears, click More info → Run anyway.

Note: No installer is required. The application is a single executable.

2.2 Linking Your Results Directory

On first launch, the app shows a prompt to link your FinishLynx results directory. This is the folder where FinishLynx writes .lif result files.

8. Click Select Results Folder.
9. Navigate to and select your FinishLynx results directory.
10. Click OK / Open.

The app will immediately start monitoring the directory. Any .lif files already present will be loaded, and new files will be detected automatically as races finish.

Tip: The results directory is typically the same folder configured in FinishLynx under Settings → Result Directory.

Note: The selected directory is remembered between sessions. Use Change Folder to switch to a different directory.

2.3 Connecting to FinishLynx

PolyField Track receives live timing data from FinishLynx via two channels: automatic results file detection and the live UDP clock feed.

Channel 1 — Results Files (.lif)

No configuration is required for results. PolyField Track monitors the results directory and automatically detects and loads new .lif files the moment FinishLynx writes them. All existing .lif files are loaded when the app starts.

FinishLynx writes a .lif file after each race. The file contains:

- The event name and wind reading
- One row per finisher: place, bib/ID, first name, last name, club, and time
- DQ and DNF entries, which are displayed at the bottom of the results without a place number
- DNS entries are silently excluded from the display

Tip: Configure FinishLynx under Settings → Result Directory to write .lif files to your chosen results folder. The same folder is used for EVT files, JPG images, and any Timetronics output.

Channel 2 — Live Clock & Start List (UDP Port 5001)

PolyField Track listens on UDP port 5001 for FinishLynx LSS (Live Scoreboard System) broadcasts. The LSS feed carries three types of data:

Data	Description
Running clock	Real-time timing clock state: armed (ready), running (counting), stopped (photocell triggered). Drives the Running Clock and Clock Overlay widgets.
Event name	The name of the event currently loaded in FinishLynx. Drives the Event Name (Current) widget and is used to look up athletes in lynx.evt.
Wind reading	Live wind gauge reading in m/s. Drives the Wind (Current) widget.

To enable the LSS feed in FinishLynx:

11. In FinishLynx, open Options → Scoreboard.
12. Click New to add a scoreboard.
13. Set Type to Network and Protocol to LSS.
14. Set IP Address to the address displayed in the dashboard (e.g. 192.168.1.255) or 127.0.0.1 for single-machine setups.
15. Set Port to 5001.
16. Ensure the scoreboard is enabled and save.

Script: polyfield.lss
Code Set: Single Byte
Serial Port: Network (UDP)
Port: 5001
IP Address: xxx.xxx.xxx.xxx
Running time: Normal / Options: Send results if armed ticked
Results: Auto
Time Precision: Hundreths

Tip: Once the LSS feed is active, the Running Clock and Event Name (Current) widgets update in real time as each race is timed. Without LSS, only static results from .lif files are available.

Note: If you are unsure of your network broadcast address, use 127.0.0.1 when FinishLynx and PolyField Track are on the same machine, or ask your network administrator for the subnet broadcast address.

Channel 3 — Start List (lynx.evt)

PolyField Track reads the FinishLynx event file (lynx.evt) to populate the Start List widget and to identify WPA-classified para-athletes for RAZA scoring.

- lynx.evt must be placed in the same results directory as your .lif files. FinishLynx can be configured to export it there automatically.
- When FinishLynx broadcasts an event name via LSS, PolyField Track looks up that event in lynx.evt to find the athlete list for the Start List widget.
- If no LSS start-list broadcast has been received (e.g. FinishLynx has not loaded an event yet), the Start List widget falls back to the first event defined in lynx.evt.
- lynx.evt is monitored for changes. If you modify or regenerate it during a meeting, PolyField Track reloads it automatically within seconds.
- WPA class codes in column 8 of lynx.evt (format: WPA 34-M, WPA 54-W) are used to identify para-athletes for RAZA scoring. See Section 13.5 and Chapter 8.2.4 for full details.

Tip: Export lynx.evt from your FinishLynx schedule before the competition begins. Place it in the results directory. PolyField Track will automatically keep start lists current throughout the day as events are updated.

2.4 Connecting to Timetronics

PolyField Track supports Timetronics timing systems in addition to FinishLynx. Timetronics writes results in two formats that PolyField Track detects and displays automatically:

File extension	Description
.res	Timetronics standard results export. TAB-delimited. Contains place, lane, time, ID, name, and club.
.txt	Timetronics text results export. Same structure as .res but with some field differences (see below).

Setting Up Timetronics Output

Configure Timetronics to write result files to the same directory that PolyField Track is monitoring:

17. In Timetronics, open the results export settings.
18. Set the output directory to the same folder selected in PolyField Track.
19. Enable continuous results export so files are written as each race finishes.

PolyField Track detects new .res and .txt files in the monitored directory the same way it detects .lif files. No additional configuration is required in PolyField Track.

What Is Available with Timetronics

Timetronics results files provide the same core data as FinishLynx .lif files. However, because Timetronics does not support the LSS UDP protocol, some features are not available:

Feature	Available	Notes
Results display	✓ Yes	Place, name, time, and ID from .res/.txt files displayed on all screens.

Event name	✓ Yes	Read from the filename or # Event: comment in the .res/.txt file.
Wind reading	✓ Yes	Read from the header row of .res files when present. Not available in .txt files (N/A is stripped).
Club / affiliation	✓ Yes (.res)	Available in .res files. Not available in .txt files (field 5 in .txt is an information field, not club name).
Running clock (live)	✗ No	Timetronics does not broadcast a UDP clock feed. Running Clock widget shows Time of Day when no LSS is active.
Start List widget	✗ No	Start list data requires lynx.evt from FinishLynx. Not produced by Timetronics.
RAZA scoring	✗ No	RAZA requires WPA class codes from lynx.evt, which is a FinishLynx file.

Note: If your meeting uses Timetronics for timing but also has access to a FinishLynx system for photo finish review, you can use both simultaneously. Place both .lif and .res files in the same directory — PolyField Track will display the most recently modified file from either system.

Event Names in Timetronics Files

PolyField Track reads the event name from Timetronics files in two ways:

- From the filename: the filename without its extension is used as the event name (e.g. a file named "100m-Men-Final.res" displays as "100m-Men-Final").
- From a comment line: if the file contains a line starting with # Event: the text after the colon is used as the event name instead of the filename (e.g. "# Event: Men's 100m Final").

Tip: If your Timetronics configuration produces filenames that are not suitable for display (e.g. containing internal codes), you can add a # Event: line to the export template to provide a clean event name.

DNS, DQ and DNF Handling

PolyField Track handles special results consistently across all file formats:

Code	Meaning	How displayed
DNS	Did Not Start	Entry is silently excluded from the display. DNS athletes do not appear in results.
DNF	Did Not Finish	Displayed at the bottom of the results list without a place number.
DQ	Disqualified	Displayed at the bottom of the results list without a place number.

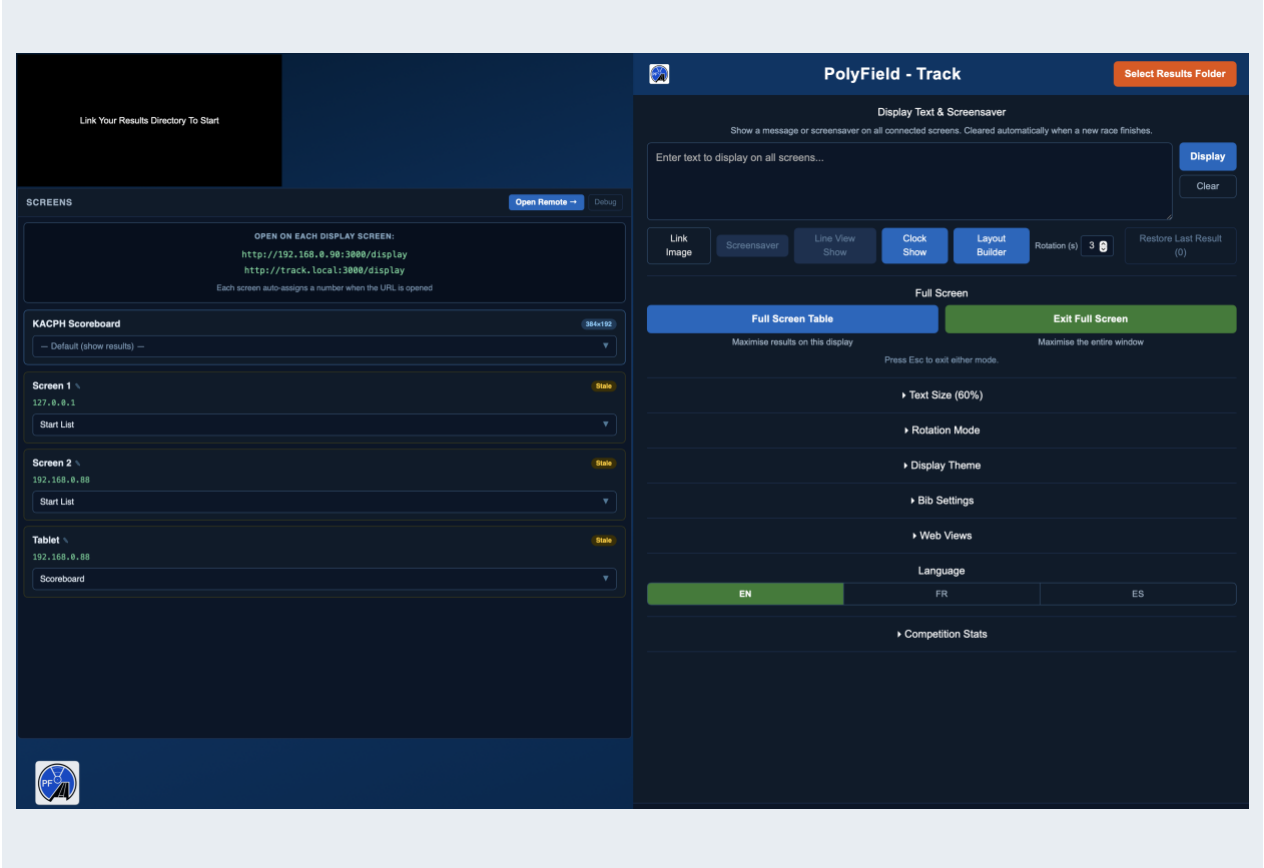
2.5 Verifying the Connection

Once set up, you can verify everything is working:

Check	How to verify
Directory linked	App header shows "Monitoring directory for new results"
Results loading	Latest result shown in top-left preview window
UDP clock active	Running Clock widget shows live time during a race (FinishLynx only)
Screens accessible	Navigate to http://track.local:3000 on another device
Display screen working	Screen registers and appears in the Screens panel
lynx.evt loaded	Start List widget populates when an event is loaded in FinishLynx

3 The Desktop Control Panel

The desktop app is the operator's primary control interface. It is divided into three main zones.



3.1 Panel Overview

Zone	Purpose
Top-left preview	A 384 × 192 px live preview of the current display output. Pixel-to-pixel mapped to the KACPH scoreboard hardware.
Left control panel	All operator controls: display mode, text, screensaver, clock, settings, competition stats, and web views.
Right screens panel	Screen connection status, layout assignment, and the KACPH Scoreboard configuration.

3.2 Top-Left Preview Window

The 384 × 192 pixel preview in the top-left corner of the app window shows exactly what the display screens are showing. It updates in real time as results change or the operator switches modes.

3.3 Right Control Panel

The right panel is divided into collapsible sections. Click any section header to expand or collapse it.

Directory & Navigation

- **Opens a file picker to choose the FinishLynx results directory:** Select Results Folder / Change Folder
- **Reopens the previously displayed result if a new race has cleared the display:** Restore Last Result
- **Opens the Layout Builder for designing custom display layouts:** Layout Builder

Display Text & Screensaver

Use these controls to push overlay content to all screens.

- **Type a message to display on all screens:** Text input field
- **Sends the typed text to all screens immediately:** Display button
- **Removes any displayed text or screensaver:** Clear button
- **Links a JPG or PNG file to use as screensaver:** Link Image / Change Image
- **Pushes the linked image to all screens as a full-screen overlay:** Screensaver button

> **Note:** Text and screensaver overlays are cleared automatically when a new race result file arrives.

Text Size

Adjusts the default text size for all displays.

- **Decrease or increase text multiplier in 5% steps:** – / + buttons

Range: 5% to 200%. The current percentage is shown between the buttons. Affects all results table text on standard display screens (not custom layout widgets).

Rotation Mode

Controls how results with more competitors than fit on screen are displayed.

Mode	Behaviour
Scroll	Top 3 positions are fixed; positions 4+ scroll continuously. Best for sprint events with a clear winner.
Page	Shows pages of up to 8 competitors, advancing automatically. Best for field events and long results lists.
Scroll All	All positions scroll through continuously. Best for events where every time matters equally.

Display Theme

Sets the visual colour scheme for all displays simultaneously.

Theme	Description
Classic	Deep navy and blue. Traditional athletics timing board aesthetic.
Modern Dark	Dark background with light text and gold highlights. Clean contemporary look.
Light	Light grey background with dark text. Professional, print-inspired.
High Contrast	Maximum contrast for visibility in bright outdoor conditions.

Bib Settings

- **Displays bib/ID numbers in results tables (default):** Show Bibs
- **Removes bib column from all displays — useful if bibs are not assigned:** Hide Bibs

Clock Controls

- **Pushes a full-screen clock overlay to all screens. The clock runs from the FinishLynx UDP feed.:** Clock Show button
- **Removes the clock overlay from all screens:** Clock Hide button

> **Note:** The Clock Overlay widget must be present in an active custom layout for this to take effect on custom layout screens. For standard display screens it always works.

Line View

Line View alternates between the FinishLynx photo finish image and the current results display.

- **Activates or deactivates the alternating line view mode:** Line View button (toggle)

The app cycles between the JPG photo finish image and the results table on a timed interval. This is useful while the photo finish is being reviewed in FinishLynx.

> **Note:** Line View requires JPG files to be present in the monitored results directory. FinishLynx writes these automatically.

Full Screen

- **Maximises the entire PolyField Track window to fill the screen:** Full Screen App
- **Expands just the results table to fill the app window (useful for the operator's own monitor):** Full Screen Table

Press Esc to exit either full-screen mode.

Web Views

Opens web-based display and analytics pages in the default browser.

- **Opens /results — all events in a scrolling grid:** Multi Result Mode
- **Opens /athlete — search for an individual athlete's results across all events:** Athlete Search
- **Opens the competition statistics and speed graphic generator:** Social Graphic

Competition Statistics

A read-only summary updated automatically as results arrive.

Statistic	Description
Total Distance	Sum of all race distances (in metres, kilometres, or miles) competed so far
Race Entries	Total number of individual competitor results across all events
Total Race Time	Sum of all finishing times across all events
Average Wind	Mean wind reading across all wind-reading events

Language

Sets the display language for both the desktop app UI and all web display pages.

- **English (default):** EN

- **French:** FR
- **Spanish:** ES

> **Note:** The language setting is sent to display screens and takes effect within one polling cycle (about 3 seconds).

Edit Club List

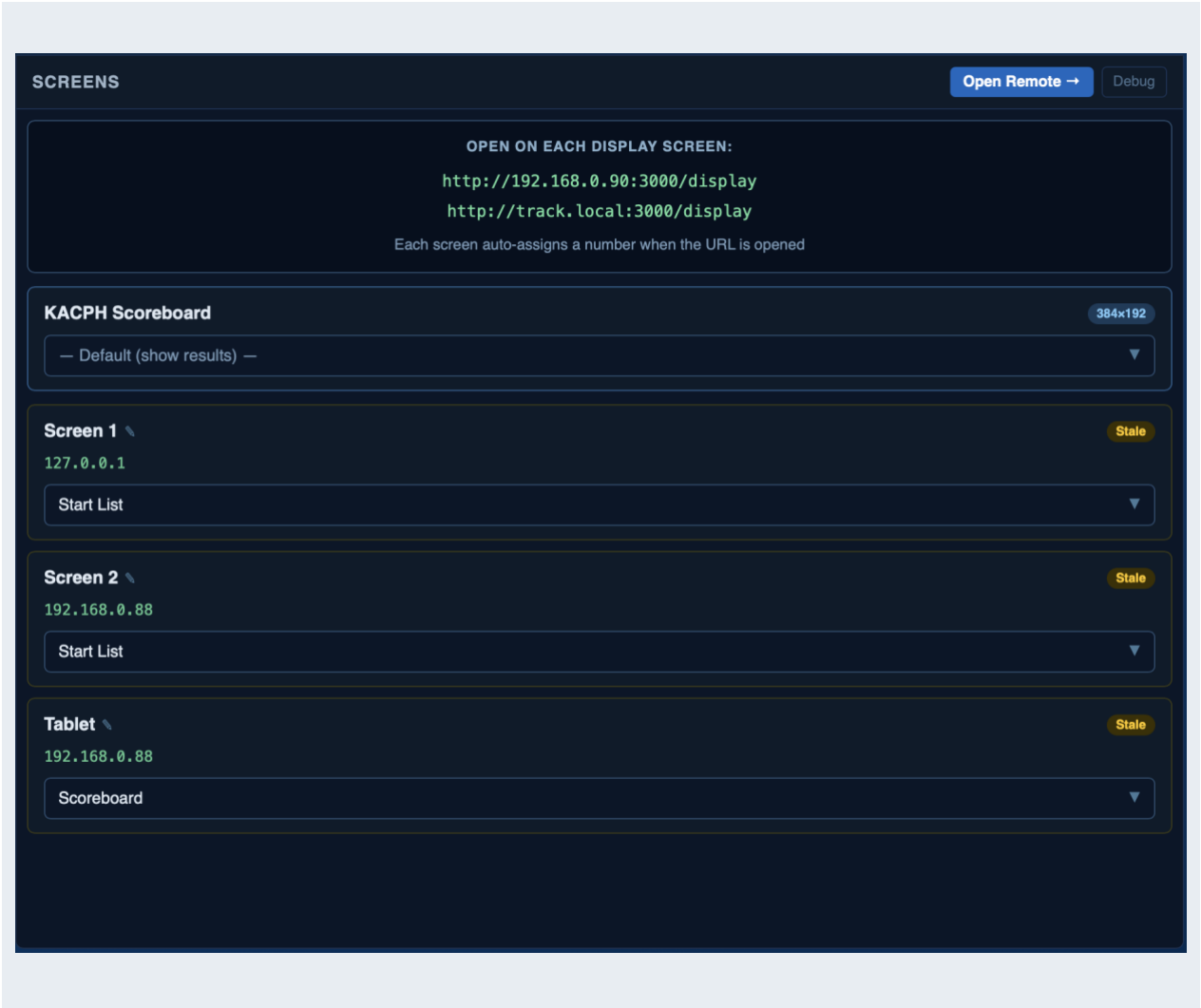
Opens club-list.csv from the results directory for editing. This file maps full club names to short abbreviations shown in results tables. See Chapter 11 for details.

Debug Log

Toggles an in-app log panel showing timestamped events: directory changes, mode switches, file detections, and server activity. Useful for diagnosing connection and display issues.

3.4 Left Screens Panel

Shows all connected display screens and the KACPH Scoreboard entry. Detailed in Chapter 5.



4 Display Modes

PolyField Track has six display modes. The current mode applies to the preview window and all connected display screens that are not assigned a custom layout.

4.1 LIF — Live Results (Default)

Shows the most recently loaded results file. This is the default mode and the one used most during competition.

The display automatically updates whenever FinishLynx writes a new .lif file. The transition is seamless — the display switches to the new result within approximately 1 second of the file being written.

100m Wheelchair Final				+0.3 m/s
1	Marcel Hug	101	SUI	13.63
2	Daniel Romanchuk	102	USA	13.79
4	Jetze Plat	103	NED	13.97
3	Nathan Maguire	201	GBR	14.11
5	David Weir	104	GBR	14.29
7	Tomoki Suzuki	105	JPN	14.42
6	Yusuke Sugawara	202	JPN	15.15
8	Rick Mitifu	213	RSA	15.88

- Rotation mode controls how large result sets are presented (see 3.3)
- Text size multiplier controls font scaling (see 3.3)
- Theme controls the colour scheme (see 3.3)

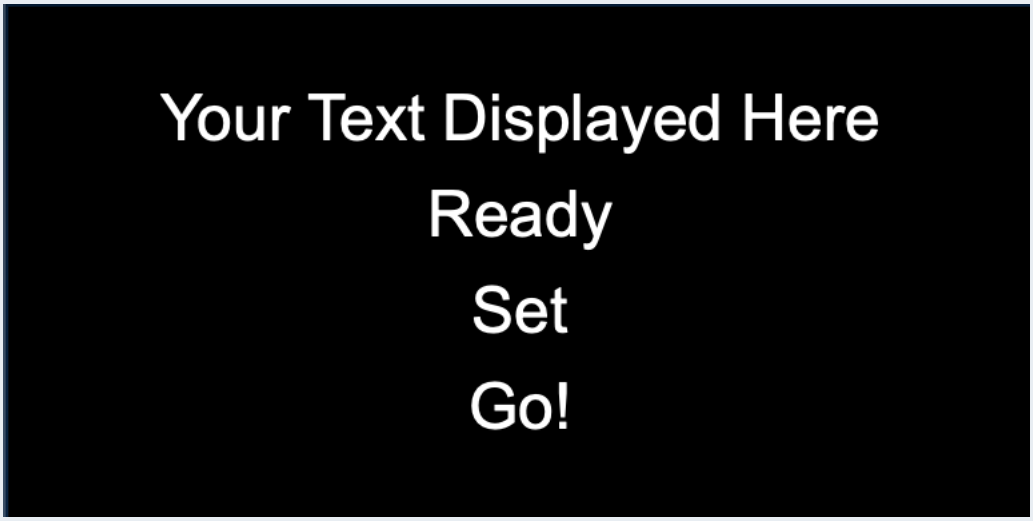
Tip: Use Restore Last Result to go back to the previous event's results if needed — useful if a new event was loaded accidentally.

4.2 Text — Custom Announcement

Displays a full-screen text message on all connected screens.

20. Type your message in the text input field in the Display Text & Screensaver section.
21. Click Display.

The text appears on all screens immediately. It is cleared automatically when the next race result arrives, or manually by clicking Clear.



Use for: Athlete announcements, schedule changes, sponsor messages, countdowns, safety notices, or any real-time communication.

4.3 Screensaver — Background Image

Displays a full-screen image on all connected screens.

22. Click Link Image to select a JPG or PNG file.
23. Click Screensaver to push it to all screens.

The image fills the screen using object-fit contain — the full image is always visible. Areas outside the image are transparent, allowing any custom layout widgets behind it to remain visible.

Use for: Sponsor logos, venue branding, holding graphics, event programmes, or any static image shown between races.

4.4 Line View — Photo Finish Alternation

Alternates between the FinishLynx photo finish image (JPG) and the current results table. Designed to be used immediately after a close race while the operator reviews the finish.

24. Ensure FinishLynx is writing JPG files to the monitored directory (it does this automatically if configured).
25. Click Line View to activate.

The display cycles: JPG image → results table → JPG image → ...

The JPG shown is the most recently modified JPG file in the results directory. The cycling interval is set by the rotation mode timer.

Tip: Click Line View again to return to standard LIF results mode.

4.5 Clock — Full-Screen Timing Clock

Displays the FinishLynx running clock full-screen on all connected screens. The clock is driven entirely by the UDP LSS feed from FinishLynx.

State	Colour	Display	Meaning
Armed	Grey	READY	Gun is set and waiting for start signal
Running	Blue	Time	Clock counting up from gun

Stopped	White	Time	Photocell triggered, final time shown
Time of Day	White	HH:MM:SS	No active event — shows current system time

- Use Clock Show button to activate the clock overlay on all screens
- Use Clock Hide to remove it and return to results

Note: On custom layout screens, the Clock Overlay widget must be in the active layout for the clock controls to affect that screen.

4.6 Custom Layout Mode

When a custom layout is activated in the Layout Builder, display screens assigned to that layout switch to it and render all its widgets with live data.

Custom layouts can combine results, clocks, logos, event names, wind, start lists, and other widgets in any arrangement. See Chapter 6 for the full Layout Builder guide.

5 Screen Setup & Management

5.1 Connecting a Display Screen

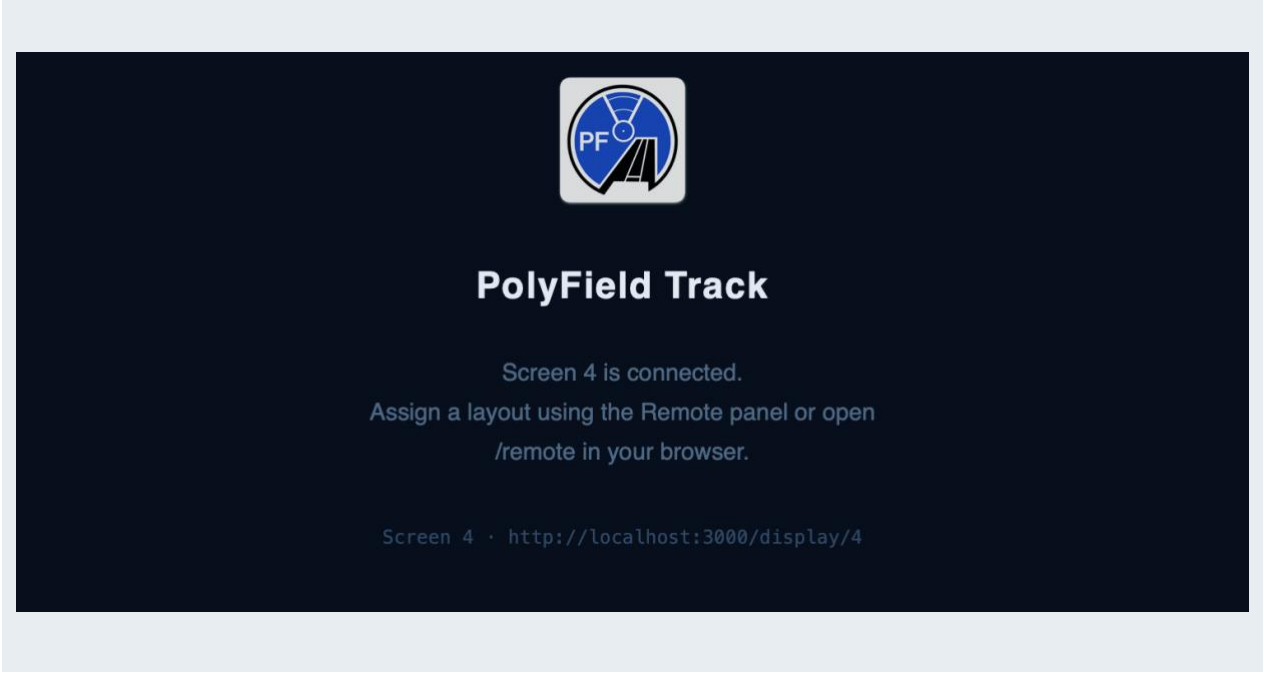
Any device with a modern web browser can become a display screen.

26. Ensure the display device is on the same local network as the timing computer.
27. Open a web browser on the display device.
28. Navigate to one of these URLs:

    `http://track.local:3000/display`

    `http://[server-IP]:3000/display`

The screen automatically registers with PolyField Track and is assigned the next available screen number (e.g. Screen 1, Screen 2, etc.). It appears in the Screens panel on the desktop app within seconds.



> **Tip:** The server IP address is shown in the Screens panel connection info block. mDNS (track.local) works on macOS, iOS, Android, and most Linux systems. Windows users may need to use the IP address directly.

5.2 The Screens Panel

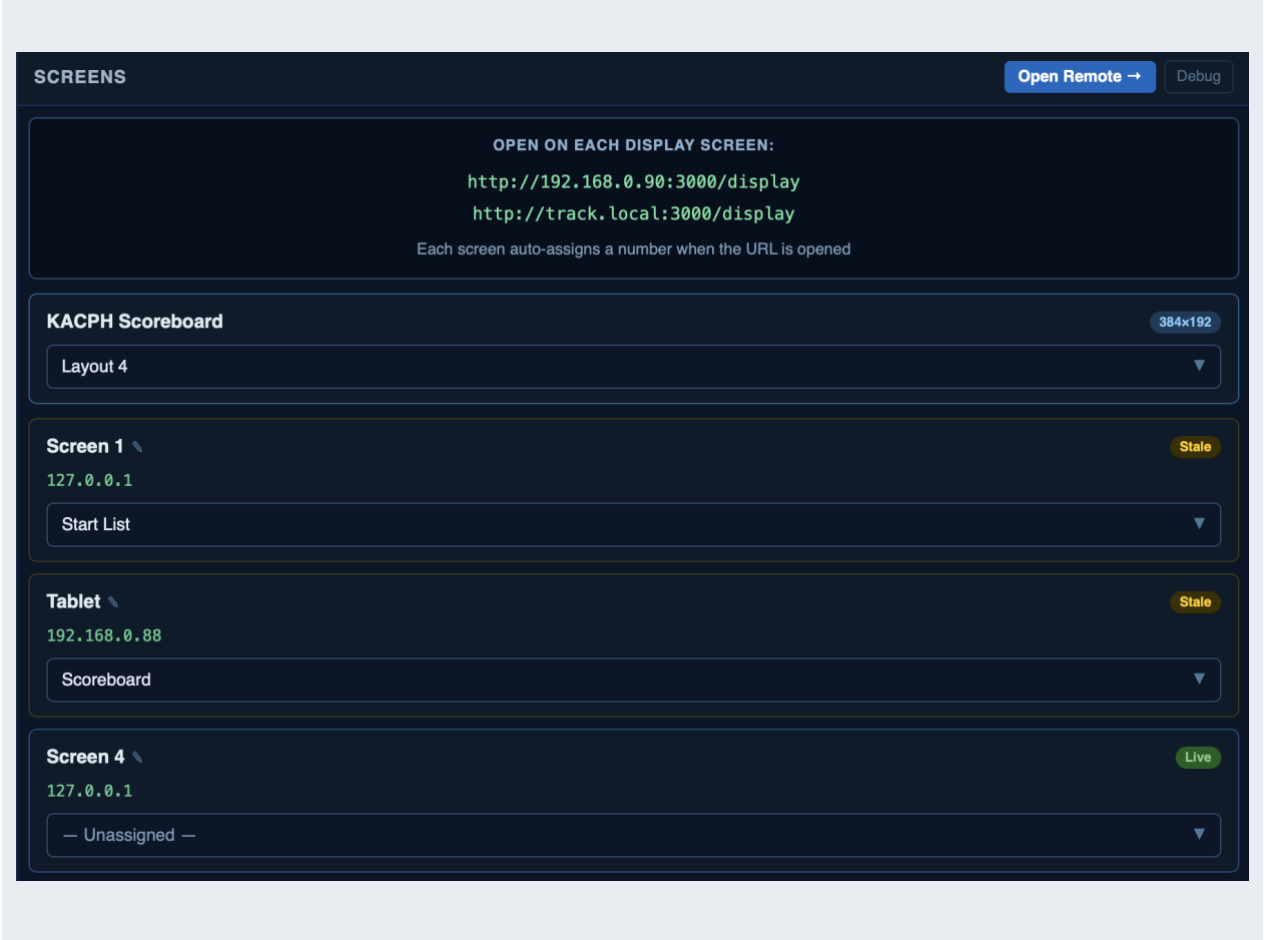
The Screens panel on the right side of the desktop app shows all currently registered screens.

Screen Cards

Each connected screen appears as a card showing:

Element	Description
Name	Editable screen name. Click the name to rename (e.g. "Back Straight",

	"Finish TV").
Status badge	Live (green) — screen connected within last 15 seconds. Stale (amber) — not recently seen.
IP address	The IP address the screen connected from.
Layout dropdown	Select which saved layout to show on that screen. Select — Unassigned — to show the default display.
✕ Remove	Removes the screen from the registry. It will re-register automatically if still open.



Assigning Layouts to Screens

29. In the Screens panel, find the screen card you want to configure.
30. Click the layout dropdown (shows "— Unassigned —" if not yet assigned).
31. Select any saved layout from the list.

The screen switches to the selected layout within 3 seconds. Each screen can show a different layout independently.

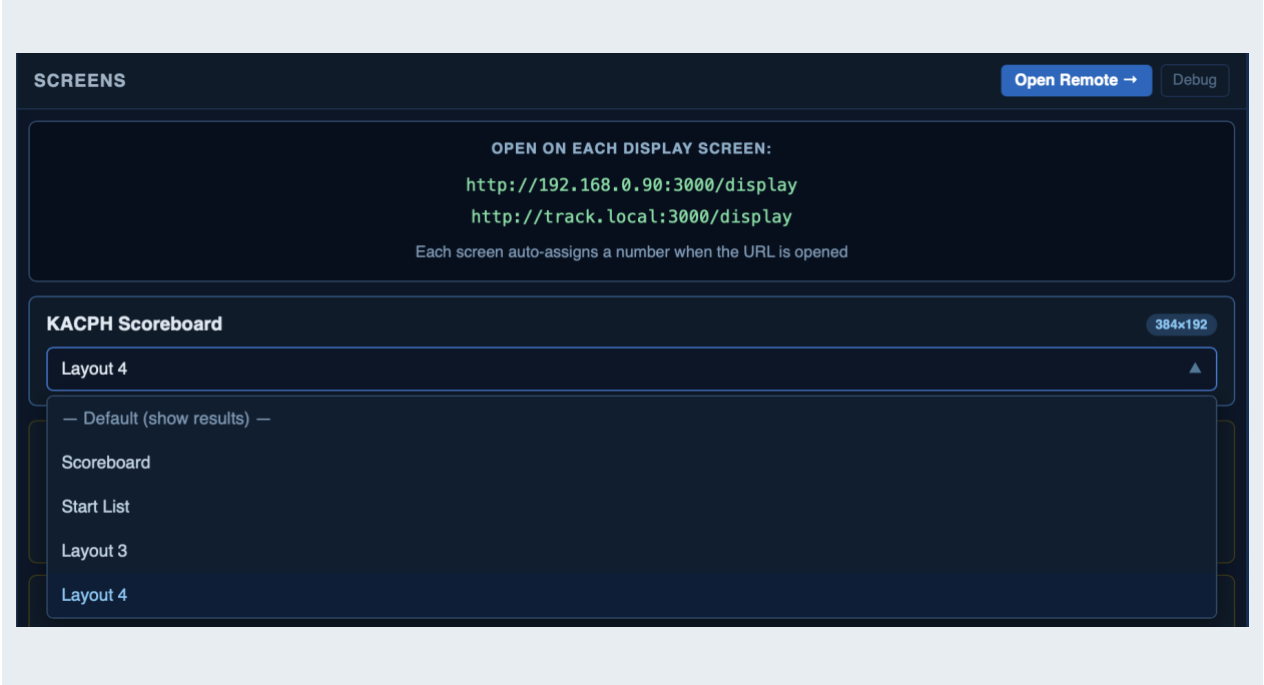
Connection Info Block

The top of the Screens panel shows the URLs to use for connecting display screens, including the LAN IP address and the track.local mDNS address.

- **Opens the /remote web panel in the default browser for tablet/phone-based screen management:** Open Remote button

5.3 KACPH Scoreboard — Desktop Preview

The KACPH Scoreboard is a permanent virtual screen in the Screens panel. It represents the 384 × 192 pixel preview window in the top-left of the desktop app, which is pixel-to-pixel mapped to the KACPH scoreboard hardware.



How It Works

The physical KACPH scoreboard hardware mirrors the exact region of the screen where the PolyField Track desktop preview window sits. Whatever appears in that 384 × 192 window is shown on the scoreboard in real time.

Layout Assignment

The KACPH Scoreboard card in the Screens panel has a layout dropdown with two options:

- **The preview window shows the normal LIF results display — the standard behaviour for scoreboard pass-through:** Default (show results)
- **Select a layout to render it in the preview at 384 × 192 px. The physical scoreboard shows that layout live.:** Any saved layout

> **Tip:** Design scoreboard layouts using the 2:1 — KACPH Scoreboard aspect ratio in the Layout Builder. This matches the 384 × 192 px dimensions exactly.

Scoreboard Web View

The dedicated /scoreboard web page shows the scoreboard-assigned layout full-screen. This can be opened on any device:

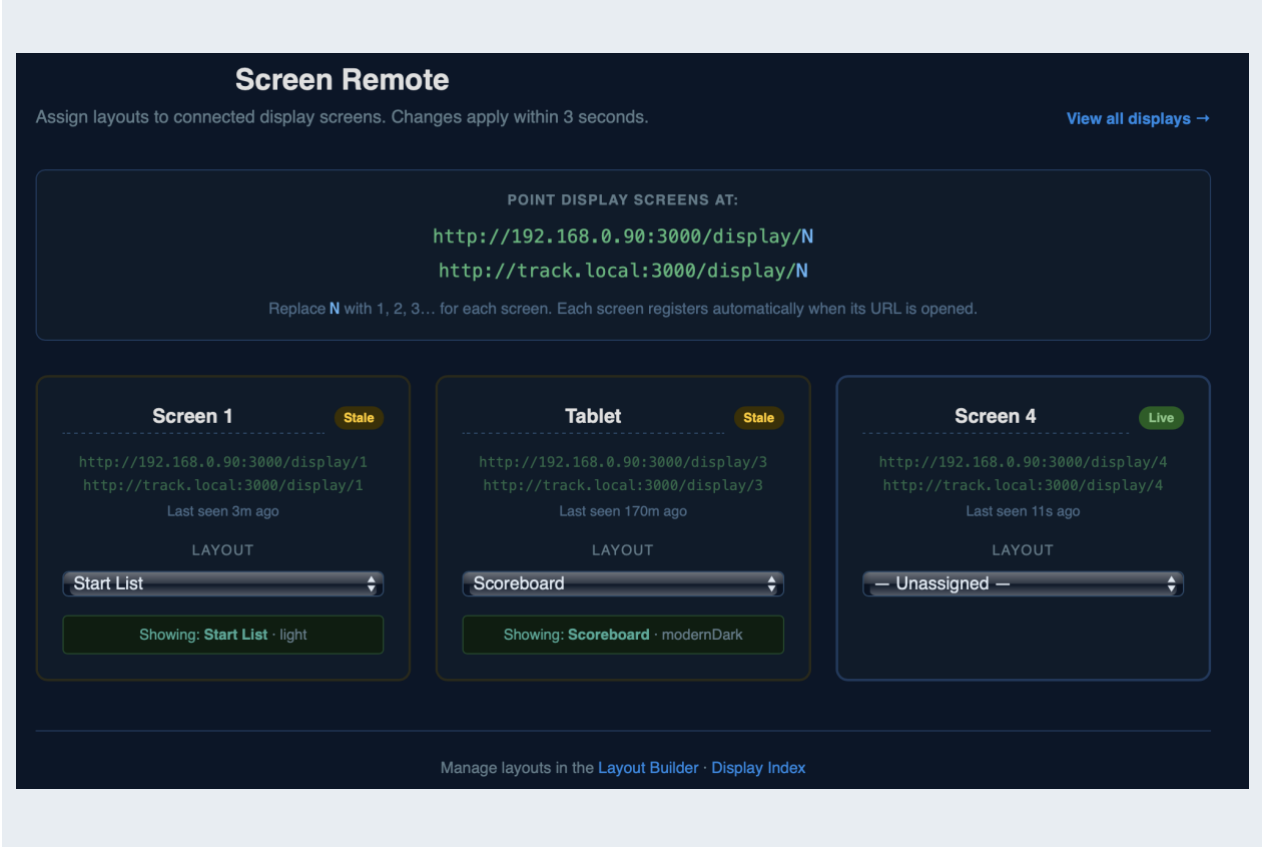
```
http://track.local:3000/scoreboard
```

The scoreboard layout is independent of the globally activated display layout — you can run different content on the scoreboard and regular screens simultaneously.

5.4 The Remote Panel (/remote)

The Remote Panel is a web page that provides screen management from any device on the network — tablet, phone, or laptop. It is particularly useful for an assistant managing screens while the operator focuses on timing.

```
http://track.local:3000/remote
```



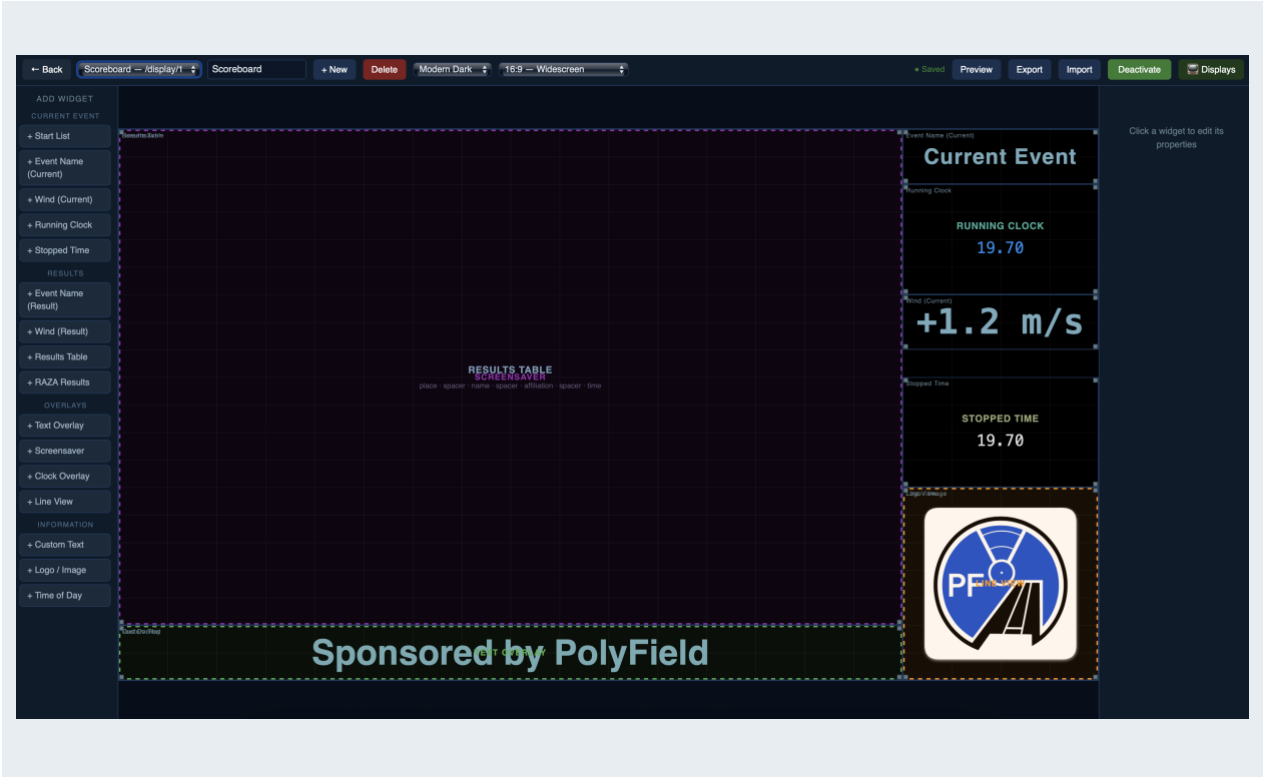
The Remote Panel shows:

- All connected screens with their current layout assignment and Live/Stale status
- Layout dropdowns to reassign screens from any device
- Screen renaming by clicking the screen name
- Last-seen timestamp (e.g. "just now", "8s ago", "2m ago")
- Connection instructions with direct screen URLs for each index

6 The Layout Builder

The Layout Builder is the design environment for creating custom display layouts. A layout is a saved arrangement of widgets on a canvas, assignable to any display screen.

Open the Layout Builder by clicking Layout Builder in the desktop app's left panel, or by navigating to /layout-builder in a browser.



6.1 Interface Overview

Area	Location	Purpose
Layout list	Left of canvas	All saved layouts. Click to select; click ⊕ New Layout to create.
Widget panel	Left sidebar	Click any widget to add it to the active canvas.
Canvas	Centre	Drag, resize, and position widgets visually.
Properties panel	Right sidebar	Configure the selected widget's options.
Toolbar	Top of canvas	Aspect ratio, theme, activate/deactivate, export/import.

6.2 Creating Your First Layout

32. Click New Layout. Enter a name.
33. Choose an aspect ratio from the toolbar (16:9 for most TVs).
34. Click a widget in the left panel to add it to the canvas.
35. Drag the widget to position it. Drag edges or corners to resize.
36. Select the widget and configure it in the Properties panel.

37. Layouts save automatically. A "Saved" indicator confirms each save.
38. Click Activate to make the layout live on all screens assigned to it.

> **Tip:** Design layouts at the aspect ratio of your target screens. A 16:9 layout will look best on widescreen TVs.

6.3 Widget Groups & Adding Widgets

Widgets are grouped by type in the left panel. Click any widget to add it to the canvas at its default size.

Group	Data source	Widgets
Current Event	Live data from FinishLynx UDP feed	Start List, Event Name (Current), Wind (Current), Running Clock, Stopped Time
Results	Data from the most recent .lif results file	Event Name (Result), Wind (Result), Results Table, RAZA Results
Overlays	Operator-triggered full-screen content	Text Overlay, Screensaver, Clock Overlay, Line View
Information	Static or system-generated content	Custom Text, Logo/Image, Time of Day
Layout Tools	Canvas configuration aids	Overlay Area (Mask)

6.4 Positioning and Resizing

The canvas uses a 20 × 20 grid. Widgets snap to grid cells.

- Drag a widget to move it
- Drag any edge or corner handle to resize
- Widgets spanning the full canvas width occupy 20 grid columns

> **Tip:** Because the grid uses percentages, layouts designed at any physical size will display correctly at any screen resolution. A 16:9 layout shows identically on a 1080p TV and a 4K display.

6.5 Selecting Overlapping Widgets

When multiple widgets overlap, clicking selects the topmost widget. To select a widget underneath:

39. Click once to select the topmost widget at that position.
40. Click again in the same spot. The selection cycles to the next widget beneath.
41. Continue clicking to cycle through all widgets at that point.

The selected widget is highlighted and its properties appear in the right panel.

6.6 Layer Order

Widget stacking order is controlled from the Properties panel when a widget is selected.

Button	Effect
Bring to Front	Move the selected widget above all others
Step Fwd	Move the selected widget one position higher
Step Back	Move the selected widget one position lower
Send to Back	Move the selected widget below all others

Important: Overlay widgets (Text Overlay, Screensaver, Clock Overlay, Line View) always render above all regular widgets regardless of their layer position. This is intentional so overlays are never obscured.

6.7 Visibility Conditions

Every widget can have conditions controlling when it is visible. This allows a single layout to adapt automatically to race state — showing a start list before the race, a clock during, and results after — without the operator switching layouts.

Conditions are set in the Properties panel (Conditions section) for each widget. All conditions must be true simultaneously for the widget to appear.

Condition	When widget is visible
Always	Always visible (default)
Has wind	Only when the current event has a wind reading
Has club	Only when at least one competitor has affiliation data
Has competitors	Only when the event has competitors loaded
Clock: running	Only while the clock is counting up
Clock: stopped	Only after the photocell is triggered
Clock: armed	Only when the gun is armed and waiting
Clock: idle	Only when no event is active
Clock: NOT running	Visible in any state except running
Clock: NOT stopped	Visible in any state except stopped
Event name contains	Only if the event name includes a specified substring
Event name does NOT contain	Only if the event name does NOT include the substring

Example: Adaptive Race Layout

- Start List widget: condition Clock: armed — shows before race
- Running Clock widget: condition Clock: running — shows during race
- Results Table widget: condition Clock: stopped — shows after finish
- Wind widget: condition Has wind — shows only when wind gauge is connected

6.8 Themes and Aspect Ratios

Theme	Aspect ratio	Typical use
Classic	16:9 Widescreen	Most TVs and monitors
Modern Dark	4:3 Standard	Older monitors, projectors
Light	21:9 Ultrawide	Panoramic LED scoreboards
High Contrast	9:16 Portrait	Vertical banners and tall displays
	1:1 Square	Square LED panels
	2:1 KACPH Scoreboard ★	KACPH hardware, 384 × 192 px

6.9 Saving, Exporting and Importing

Layouts save automatically as you edit. Manual controls:

- **Downloads the layout as a .json file:** Export button
- **Loads a .json layout file from your computer:** Import button

Exported layouts include all widget configuration, positions, colours, and embedded images. They can be imported on any other machine running PolyField Track.

> **Tip:** Export layouts before major events as a backup. Share layouts between officials by exporting and emailing the .json file.

7 Widget Reference

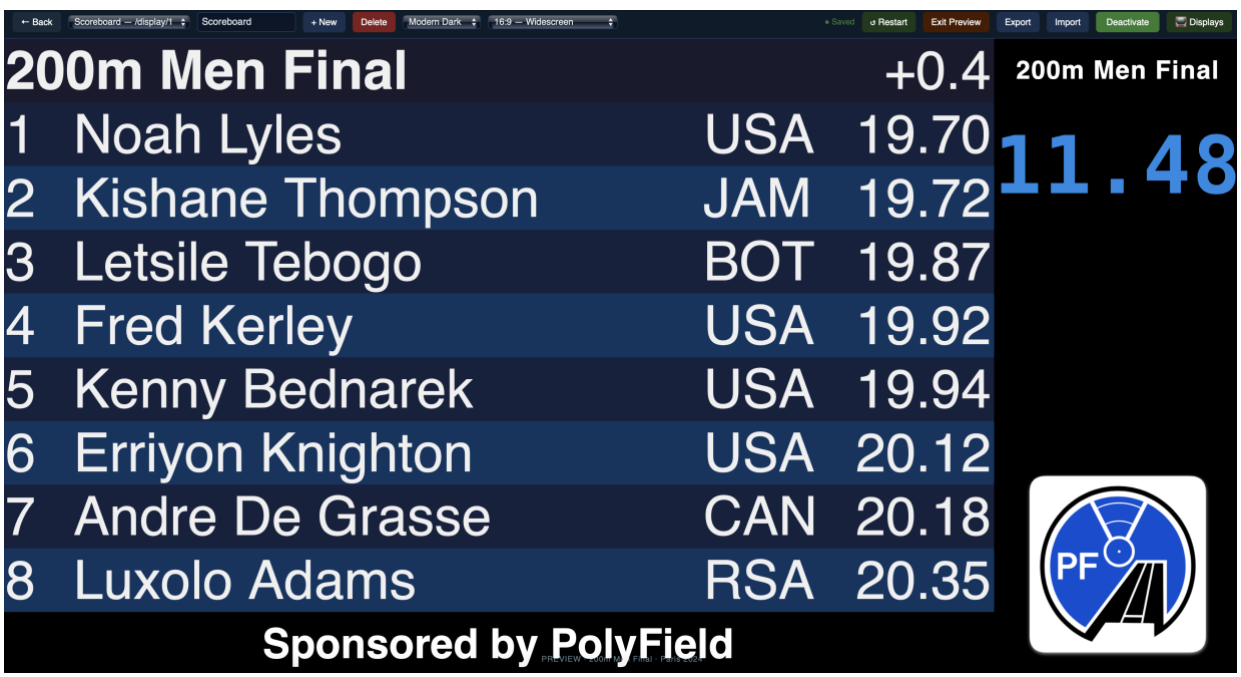
This chapter is a complete reference for every widget available in the Layout Builder. Widgets are organised into five groups matching the widget panel in the Layout Builder.

All widget positions and sizes are defined on a 20 × 20 grid. Each grid unit equals 5% of the layout canvas. Widgets snap to the grid when dragged or resized.

> **Note:** Every widget has a background colour setting. Leave it blank (transparent) to let the layout background show through. Set a solid colour to give the widget its own background box.

Group	Widgets	Data source
Current Event	Start List, Event Name (Current), Wind (Current), Running Clock, Stopped Time	FinishLynx UDP / LSS feed
Results	Event Name (Result), Wind (Result), Results Table, RAZA Results	Most recent .lif file
Overlays	Text Overlay, Screensaver, Clock Overlay, Line View, Overlay Area	Operator controls
Information	Custom Text, Logo / Image, Time of Day	Static or server time
Layout Tools	Overlay Area (Mask)	Canvas configuration

Preview the scoreboard using Preview Mode



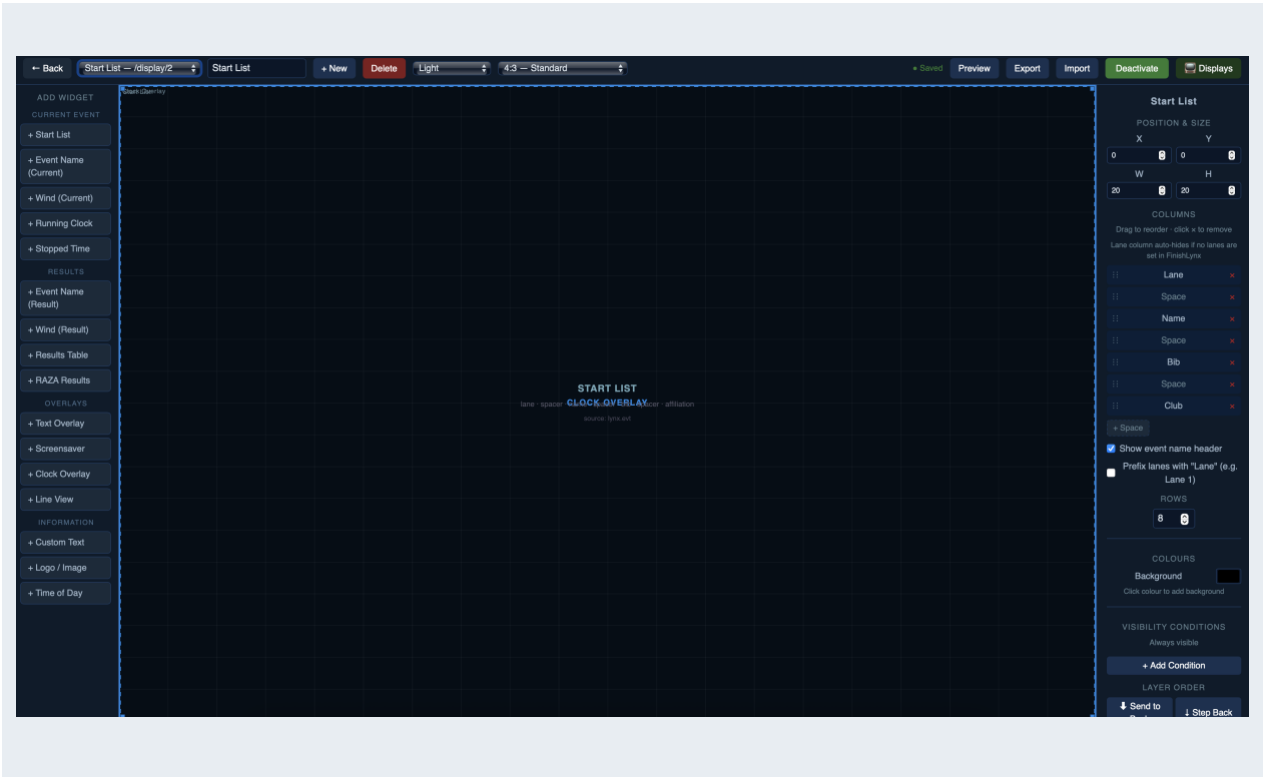
7.1 Current Event Widgets

Current Event widgets show live data from the FinishLynx UDP / LSS feed. They update in real time during a race — no polling delay. They require the LSS scoreboard to be configured in FinishLynx (see Section 2.3).

Note: If no UDP data has been received since the app started, Current Event widgets show placeholder values or the time of day until the first broadcast arrives.

7.1.1 Start List

Displays the pre-race competitor lineup read from the FinishLynx event file (lynx.evt). Updates each time FinishLynx broadcasts a new event via LSS. If no LSS broadcast has been received, it falls back to the first event in lynx.evt.



Data source: lynx.evt in the results directory, updated via LSS broadcast.

Fallback: First event in lynx.evt if no LSS start-list has been received this session.

Configurable columns:

Label	ID	Description
Lane	lane	Start lane number. Auto-hides if FinishLynx has no lane assignments.
Bib	bib	Athlete bib / ID number.
Name	name	Athlete full name (Last, First from FinishLynx).
Club	affiliation	Club or team affiliation.
Space	spacer	Empty column used to add visual separation between other columns. Multiple spacers allowed.

All other settings:

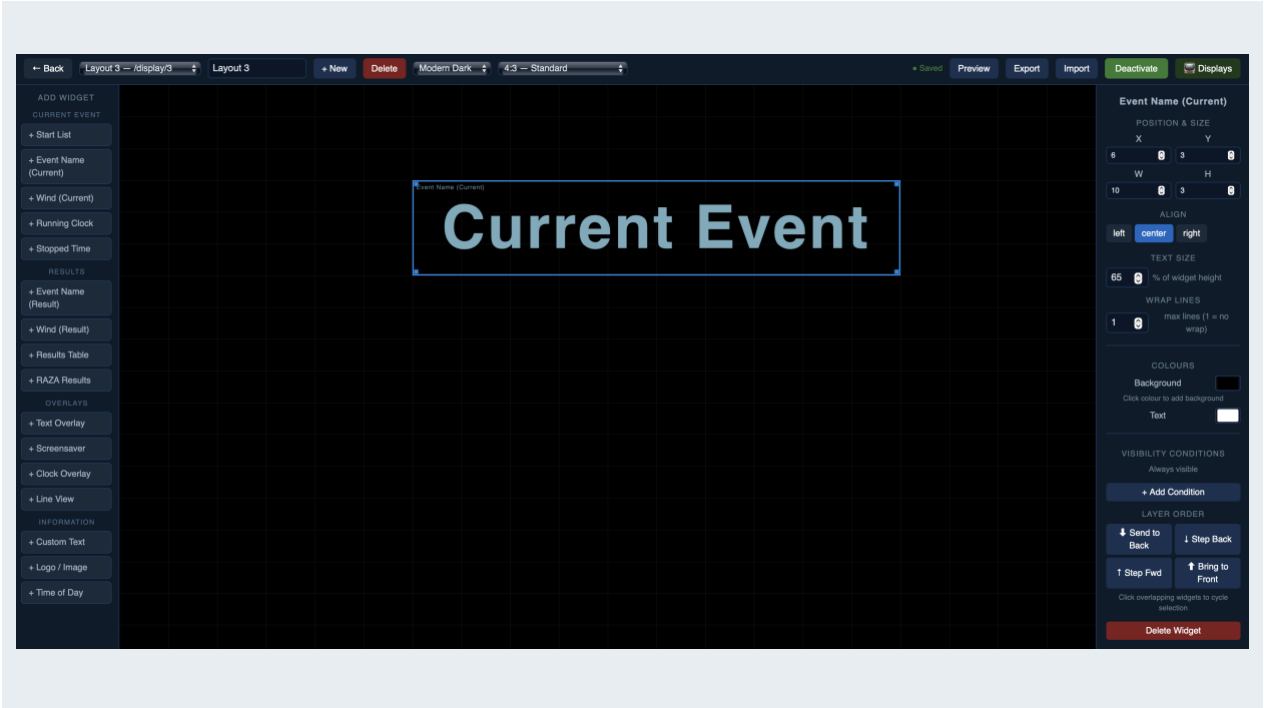
Setting	Type	Default	Description
Show event header	Checkbox	On	Displays the event name as a header row above the competitor list.
Max rows	1 – 30	8	Maximum number of competitors shown.

			Rows beyond this limit are hidden.
Lane prefix	Checkbox	Off	Prepends "Ln " before each lane number (e.g. "Ln 3" instead of "3").

> **Tip:** Add the Start List widget with visibility condition Clock: armed so it appears automatically before each race and hides once timing starts.

7.1.2 Event Name (Current)

Displays the current event name received from the FinishLynx UDP clock feed. This is the name FinishLynx is actively timing and updates the moment FinishLynx moves to a new event — before any result file is written.



Data source: FinishLynx LSS UDP clock broadcast (port 5001).

Setting	Type	Default	Description
Text align	Left / Centre / Right	Centre	Horizontal alignment of the event name text within the widget.
Text size	10 – 200%	65%	Font size as a percentage of widget height. 65% fills most of the widget height with a single line.
Wrap lines	1 – 10	1	Maximum number of lines before truncation. Set to 2 or 3 for long event names.
Text colour	Colour picker	White	Foreground text colour.
Background	Colour picker / blank	None	Widget background fill. Leave blank for transparent.

Tip: Pair with Event Name (Result) on the same position using visibility conditions: show Current when Clock: running and Result when Clock: stopped. The display then shows the live event name during timing and switches to the confirmed result name automatically.

7.1.3 Wind (Current)

Displays the live wind reading in m/s from the FinishLynx UDP clock feed. The value is the wind gauge reading for the race in progress.

Data source: FinishLynx LSS UDP wind field (port 5001).

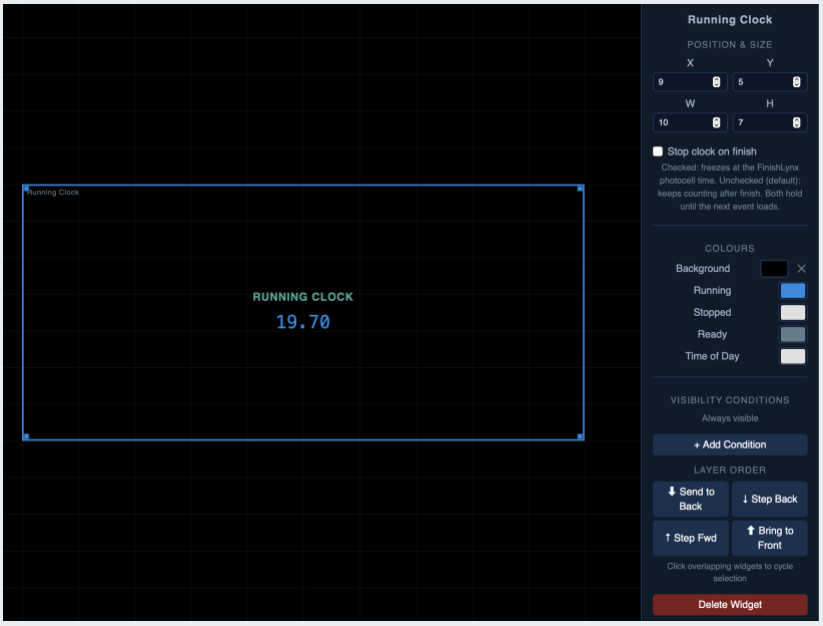
Format: Signed value with m/s suffix, e.g. "+1.8 m/s" or "−0.4 m/s".

Setting	Type	Default	Description
Text align	Left / Centre / Right	Centre	Horizontal alignment.
Text colour	Colour picker	#a0b4c8 (steel)	Wind text colour. Steel blue is the default to distinguish it from results data.
Background	Colour picker / blank	None	Widget background fill.

Tip: Use the Has wind visibility condition to hide this widget entirely for non-wind events (e.g. long distance). This prevents an empty "wind" label appearing on the scoreboard.

7.1.4 Running Clock

A live stopwatch that mirrors the FinishLynx timing clock in real time. The clock colour changes automatically to reflect the current race state.



Data source: FinishLynx LSS UDP clock feed (port 5001). Falls back to system time of day when no event is active.

Clock states and default colours:

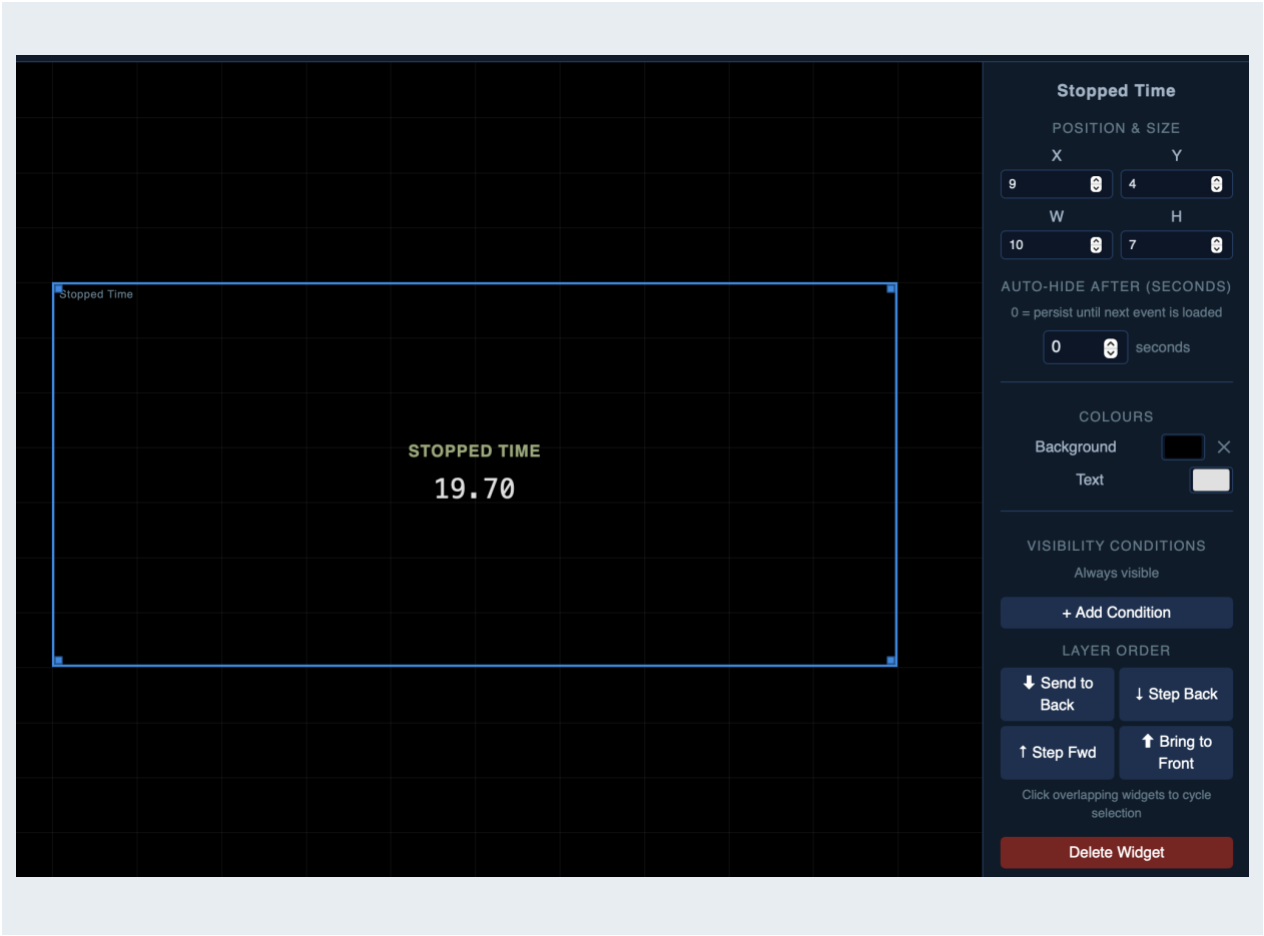
State	Default colour	Description
Armed	Grey (#607d8b)	Gun is set, waiting for start signal. Clock shows 0.00.
Running	Blue (#1e88e5)	Clock counting up from the gun signal.
Stopped	White (#e0e0e0)	Photocell triggered. Final time shown and held.
Time of Day	White (#e0e0e0)	No active event. Shows current system time (HH:MM:SS).

Setting	Type	Default	Description
Running colour	Colour picker	#1e88e5	Clock text colour when timing is active.
Stopped colour	Colour picker	#e0e0e0	Clock text colour when stopped at finish time.
Armed colour	Colour picker	#607d8b	Clock text colour in armed / ready state.
Time of Day colour	Colour picker	#e0e0e0	Clock text colour when showing wall-clock time.
Background	Colour picker / blank	Black	Widget background fill. Black is the default for maximum contrast.
Stop clock on finish	Checkbox	Off	When on: clock freezes at the photocell time. When off: clock continues counting after finish, held until next event loads.

Tip: Use Running Clock for a prominent timing display. Use Stopped Time alongside it — Running Clock shows the live time, Stopped Time shows only the confirmed final time with its own auto-hide timer.

7.1.5 Stopped Time

Shows the final photocell time from FinishLynx, held until the next event is loaded. Unlike Running Clock, this widget only shows a value after the photocell has been triggered — it is blank during armed and running states.



Data source: FinishLynx LSS UDP stopped-time field.

Visible: Only after photocell trigger. Blank during armed and running states.

Setting	Type	Default	Description
Auto-hide after	0 – 300 seconds	0 (persist)	Seconds to display the stopped time before hiding. 0 = hold until the next event loads.
Text colour	Colour picker	White	Stopped time text colour.
Background	Colour picker / blank	Black	Widget background fill.

Tip: Set auto-hide to 30 seconds in layouts that also show a Results Table. The stopped time will display briefly after the finish, then hide automatically as the results load and the table becomes the focus.

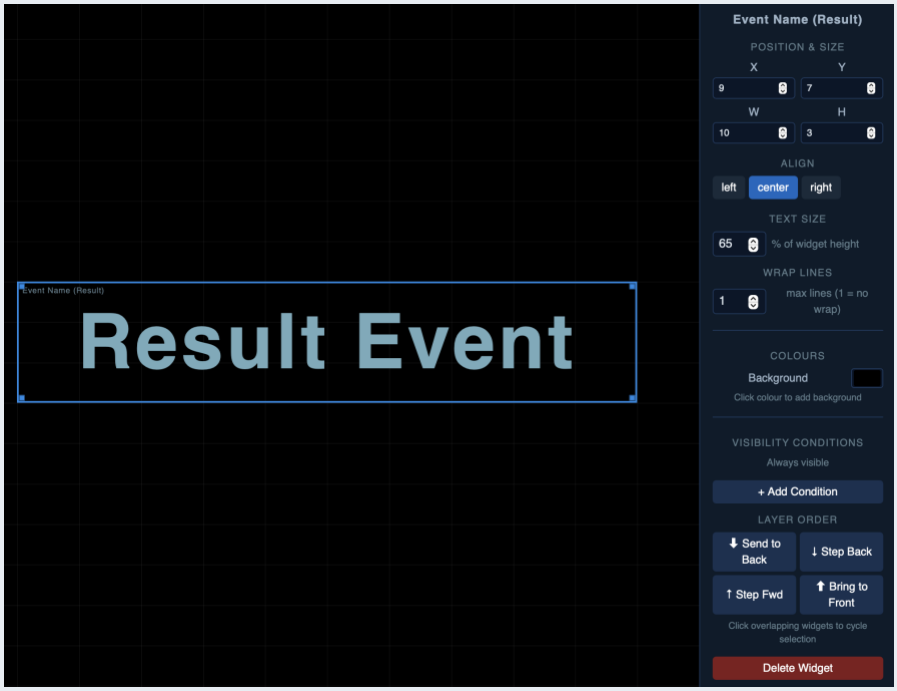
7.2 Results Widgets

Results widgets show data from the most recently loaded .lif result file. They update automatically each time FinishLynx writes a new result file to the monitored directory.

> **Note:** Results widgets are not affected by the UDP clock state. They hold the last loaded result until a new .lif file is detected, regardless of whether the clock is running or stopped.

7.2.1 Event Name (Result)

Displays the event name from the most recently loaded results file. Unlike Event Name (Current), this value does not change when FinishLynx starts timing the next race — it holds the name of the race whose results are on screen.



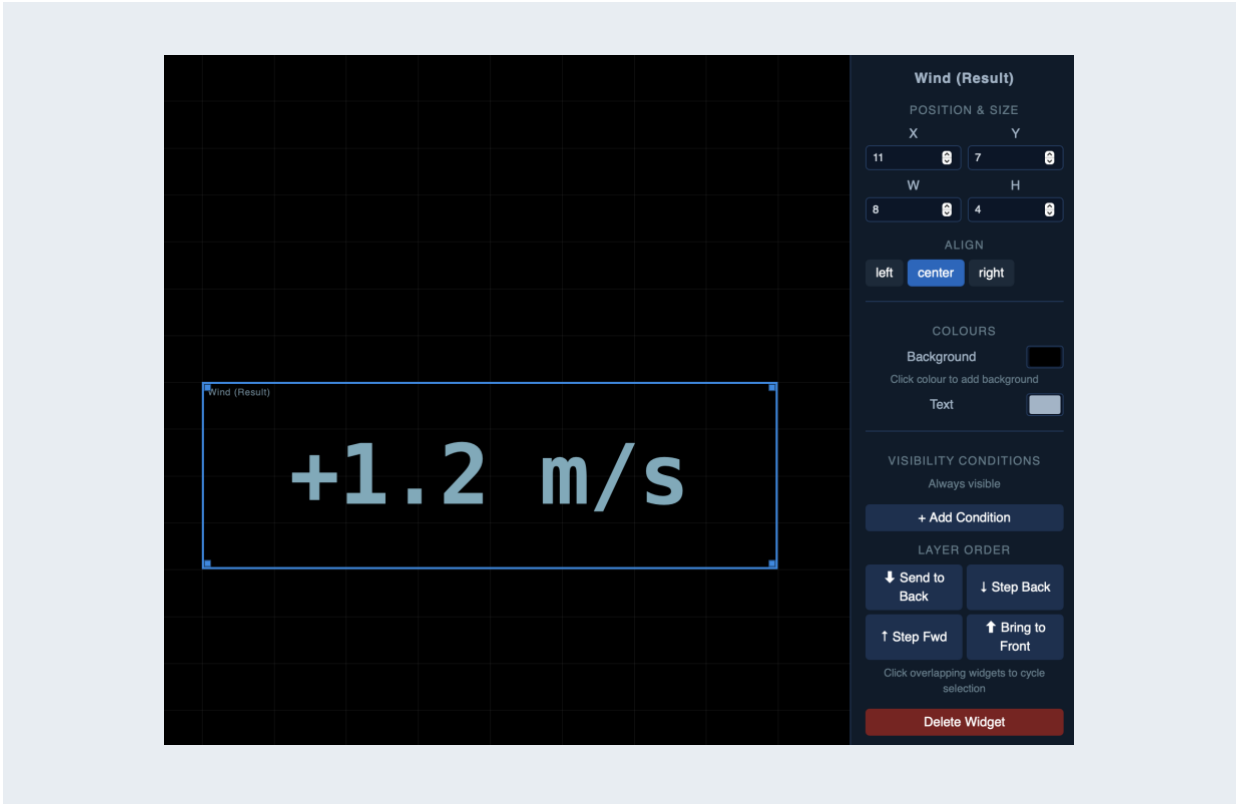
Data source: .lif results file (event header).

Use case: Pair with a Results Table to label the result on screen after the race has finished. The name persists while you are showing those results even if timing has moved to the next event.

Setting	Type	Default	Description
Text align	Left / Centre / Right	Centre	Horizontal alignment.
Text size	10 – 200%	65%	Font size as percentage of widget height.
Wrap lines	1 – 10	1	Max lines before truncation.
Text colour	Colour picker	White	Text colour.
Background	Colour picker / blank	None	Background fill.

7.2.2 Wind (Result)

Displays the official wind reading from the most recently loaded results file. This is the confirmed, recorded wind speed — not the live gauge reading.



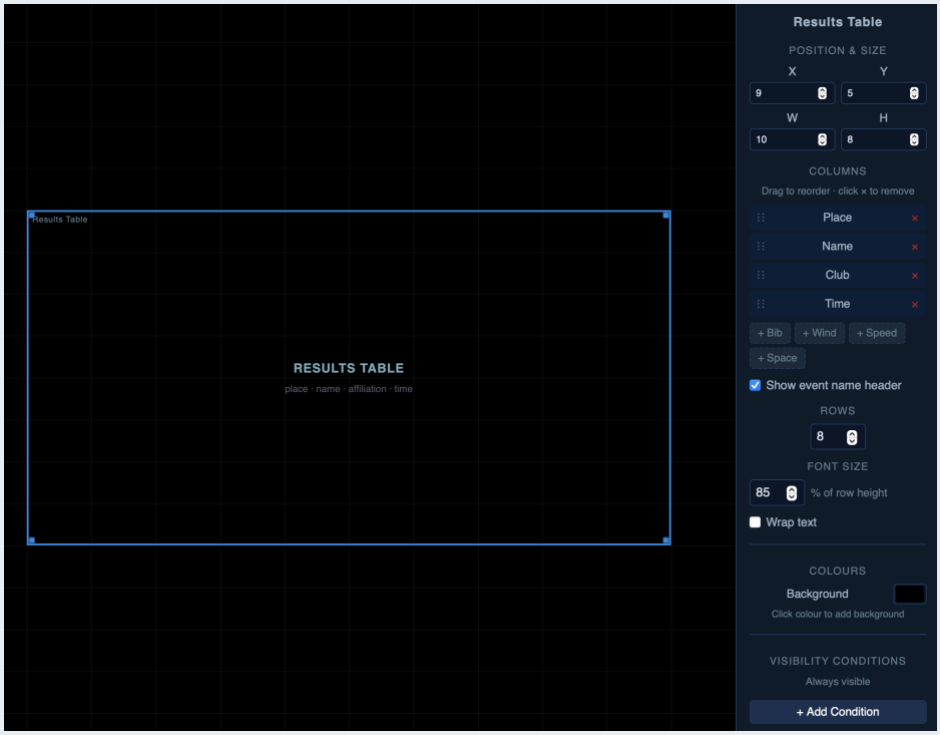
Data source: .lif results file (wind field in event header).

Setting	Type	Default	Description
Text align	Left / Centre / Right	Centre	Horizontal alignment.
Text colour	Colour picker	#a0b4c8 (steel)	Wind text colour.
Background	Colour picker / blank	None	Background fill.

> **Tip:** Use the Has wind visibility condition to automatically hide this widget for non-wind events.

7.2.3 Results Table

The primary results display widget. Shows a table of competitor results from the most recently loaded .lif file with fully configurable columns, font size, and row count. This is the main widget for showing race outcomes on any display screen.



Data source: .lif results file (all competitor rows).

Available columns (drag to reorder, click × to remove, click + to add):

Label	ID	Description
Place	place	Finishing position number (1st, 2nd, 3rd, etc.).
Bib	bib	Athlete bib / ID number.
Name	name	Athlete name (Last, First as recorded in FinishLynx).
Club	affiliation	Club or team name, substituted with abbreviation from club-list.csv if available.
Time	time	Official finishing time, including reaction time and DQ markers.
Wind	wind	Per-athlete wind reading (for combined events or wind-measured races). Hidden if no wind data.
Speed	speed	Athlete speed calculated from event distance and finishing time. See speed note below.
Space	spacer	Blank column for visual spacing. Multiple spacers can be added.

Setting	Type	Default	Description
Show event header	Checkbox	On	Displays the event name as a header row at the top of the table.
Max rows	1 – 20	8	Maximum number of result rows shown. Results beyond this limit are hidden (use rotation mode on standard displays for scrolling).
Font size	10 – 200%	85%	Row text size as a percentage of the row height. Reduce for long names; increase for large screens viewed from a distance.

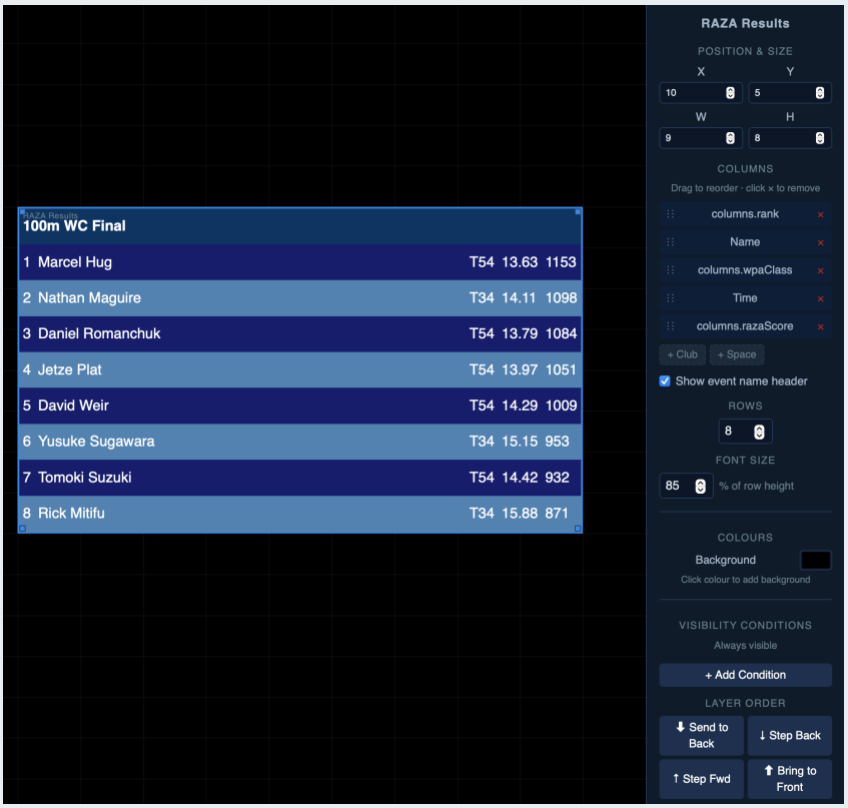
Wrap text	Checkbox	Off	When on, long names wrap to a second line within the row instead of being truncated.
Speed unit	km/h / mph / m/s	km/h	Unit used for the Speed column. Only shown when the Speed column is active.

Note: Speed calculation requires the event name to contain a recognisable distance (e.g. "100m", "200 Metres", "400m Hurdles"). Events without a recognisable distance show a dash in the Speed column.

Tip: For large scoreboard layouts where readability matters, reduce Max rows to 6 and increase Font size to 100–110%. For information screens showing full fields, increase Max rows to 16–20 and reduce font size to 70–75%.

7.2.4 RAZA Results

Shows WPA para-athletes from the current race ranked by RAZA point score, highest first. RAZA (Ranking Algorithm for Zones of Achievement) is the official World Para Athletics formula for comparing performances across different WPA classification classes.



Data source: .lif results file (for times) combined with lynx.evt (for WPA class codes).

Scoring: 2025 World Para Athletics RAZA formula. Scores are computed server-side automatically.

Prerequisite: Athletes must have a WPA class code in column 8 of lynx.evt in the format WPA · class · gender, e.g. WPA 34-M, WPA 54-W, WPA 53-V.

Available columns:

Label	ID	Description
Rank	rank	Position in the RAZA ranking (1 = highest score).
Name	name	Athlete name.
Club	affiliation	Club or team affiliation.
Class	wpaClass	WPA classification code, e.g. T34, T54, T53.
Time	time	Athlete's official finishing time from the results file.
RAZA Score	razaScore	Calculated RAZA point score (integer). Higher is better.
Space	spacer	Blank column for visual spacing.

Setting	Type	Default	Description
Show event header	Checkbox	On	Displays the event name above the table.
Max rows	1 – 20	8	Maximum rows shown.
Font size	10 – 200%	85%	Row text size as a percentage of row height.

How WPA classification codes are read from lynx.evt:

lynx.evt column 8 value	Interpreted as	Notes
WPA 34-M	T34 (Men)	WPA prefix + class number + gender suffix M
WPA 54-W	T54 (Women)	WPA prefix + class number + gender suffix W
WPA 53-V	T53 (Mixed)	WPA prefix + class number + gender suffix V
WPA 46-M	T46 (Men)	Any T-class number works; WPA prefix is required

Note: Athletes without a WPA class code in lynx.evt are not shown in the RAZA Results widget or the /raza rankings page, even if they competed in the same race.

Tip: The /raza web page (Chapter 8.5) shows a full-meeting RAZA combined ranking across all events — useful as a secondary display during para-athletics competitions.

7.3 Overlay Widgets

Overlay widgets respond to operator button presses in the desktop control panel. They always render above all other widgets regardless of their layer order in the layout. This ensures operator-triggered content is never obscured by results or other display elements.

> **Important:** Overlay widgets must be present in the active layout for the corresponding operator controls to take effect on that screen. A screen running a layout with no Text Overlay widget will not show text messages pushed from the desktop app.

Overlay activation summary:

Widget	Activated by	Cleared by
Text Overlay	Display button in desktop app	Clears automatically when next .lif result file is loaded, or when Clear is clicked.
Screensaver	Screensaver button in desktop app	Clears automatically when next .lif result file is loaded, or when Clear is clicked.
Clock Overlay	Clock Show / Clock Hide buttons	Stays active until Clock Hide is clicked. Not cleared by new results.
Line View	Line View toggle in desktop app	Cycles on a timed interval. Cleared when Line View is toggled off.

7.3.1 Text Overlay

Displays the message typed by the operator in the Display Text field of the desktop control panel. When activated, the text fills the overlay area full-screen (or within the Overlay Area mask if one is present in the layout).

Data source: Operator text input from the desktop app Display Text field.

Auto-clear: The overlay clears automatically when the next race result arrives. Click Clear to remove it manually.

Configuration: None. Appearance is controlled by the operator's display mode settings and the current theme.

> **Tip:** Use for: athlete announcements, event schedule updates, ceremony notices, safety messages, or any real-time text communication to all screens simultaneously.

7.3.2 Screensaver

Displays the image linked by the operator (JPG or PNG) as a full-screen overlay. The image is scaled to fit using object-fit contain — the full image is always visible, with transparent areas outside the image boundary. This means any layout widgets behind the screensaver remain visible in the letterbox areas.

Data source: Operator-linked JPG or PNG file from the desktop app.

Auto-clear: Clears when the next race result arrives or when Clear is clicked.

Configuration: None. The image is provided at run time by the operator, not configured in the layout.

> **Tip:** Design your layout background to look good with the screensaver active — letterbox areas show through. A dark or branded background behind the image works well.

7.3.3 Clock Overlay

Displays a full-screen timing clock triggered by the Clock Show button in the desktop app. The clock operates identically to the Running Clock widget — it mirrors the FinishLynx timing clock and changes colour based on race state — but fills the entire overlay area rather than being confined to a fixed widget region.

Activated by: Clock Show button in the desktop app left panel.

Cleared by: Clock Hide button. Not cleared by new results.

Data source: FinishLynx LSS UDP clock feed (port 5001).

Clock states:

State	Colour	Description
Armed	Default: grey #607d8b	Ready to start. Shows 0.00.
Running	Default: blue #1e88e5	Counting up from gun.
Stopped	Default: white #e0e0e0	Held at photocell time.
Time of Day	Default: white #e0e0e0	No active event. Shows wall-clock time.

Setting	Type	Default	Description
Running colour	Colour picker	#1e88e5	Clock text colour when timing is active.
Stopped colour	Colour picker	#e0e0e0	Clock text colour when stopped.
Armed colour	Colour picker	#607d8b	Clock text colour when armed.
Time of Day colour	Colour picker	#e0e0e0	Clock text colour in idle state.
Background	Colour picker / blank	Black	Overlay background colour.

> **Tip:** If your layout uses an Overlay Area (Mask) widget, the Clock Overlay will be confined to that zone rather than filling the entire screen. Position the mask widget where you want clock overlays to appear within the layout.

7.3.4 Line View

Displays the FinishLynx photo finish image (JPG) from the results directory. Designed for use immediately after a close finish while the operator reviews the image in FinishLynx. Line View alternates between the photo finish JPG and the transparent state, allowing the results table behind it to show through during the results phase.

Data source: Most recently modified .jpg or .jpeg file in the results directory.

Activated by: Line View toggle in the desktop app left panel.

Cycle: Photo finish image → transparent (results visible) → photo finish image → ... timed by the rotation interval.

Setting	Type	Default	Description
Show rotation indicator	Checkbox	Off	When on, a small progress indicator is shown during the rotation cycle so the operator knows when the image will return.

Note: Line View requires JPG files in the monitored results directory. FinishLynx writes these automatically when configured to save photo finish images. The Line View button is disabled in the desktop app if no JPG files are found.

7.3.5 Overlay Area (Mask)

An invisible positioning widget that confines all overlay content (Text Overlay, Screensaver, Clock Overlay, Line View) to a specific zone of the layout canvas. Without an Overlay Area, all overlays fill the entire screen.

Visible on display screens: No. The Overlay Area is only visible in the Layout Builder as a design guide.

Limit: Only one Overlay Area per layout.

Configuration: None. Resize and reposition it on the canvas to define the overlay zone.

When to use an Overlay Area:

- When your layout has a dedicated results zone and you want operator overlays to appear within that zone only — leaving a logo panel, clock area, or other region unaffected.
- When showing sponsor logos or event branding permanently at the edges of the screen, with the results area in the centre.
- When the KACPH Scoreboard layout needs to reserve part of the display for hardware-specific content.

Tip: In a typical scoreboard layout, place the Overlay Area over the results table region. The clock, text, and screensaver overlays will then be confined to that area, keeping the event name and logo visible at all times.

7.4 Information Widgets

Information widgets show static content defined at design time, or system-generated data that does not change during a race. They do not depend on FinishLynx connectivity.

7.4.1 Custom Text

A fixed text label set at design time. The content never changes at run time — it always shows exactly what you typed when designing the layout. Use for headings, section labels, static announcements, venue names, or any text that should be permanently visible.

Setting	Type	Default	Description
Text	Text input	Custom Text	The text to display. Can include any characters, numbers, or symbols.
Text colour	Colour picker	White	Text foreground colour.
Align	Left / Centre / Right	Centre	Horizontal text alignment within the widget.
Text size	10 – 200%	65%	Font size as percentage of widget height.
Wrap lines	1 – 10	1	Max lines (1 = no wrapping; increase for multi-line labels).
Background	Colour picker / blank	None	Widget background fill.

> **Tip:** Use Custom Text widgets as section dividers or column headers above a Results Table. For example, a "RESULTS" heading in a contrasting colour above the table helps viewers locate information quickly on a busy scoreboard.

7.4.2 Logo / Image

Embeds a static image (PNG, JPG, SVG, or GIF) into the layout. The image is stored directly in the layout file as base64 data, so no external files or network access are required when displaying the layout. Layouts can be shared between machines and the logo travels with them.

Supported formats: PNG, JPG/JPEG, SVG, GIF.

Storage: Embedded in the layout JSON file. No external file dependencies.

Setting	Type	Default	Description
Upload Image	File picker button	—	Opens a file picker to select an image. The image is immediately embedded into the layout.
Remove	Button	—	Removes the embedded image. The widget becomes blank.
Fit	contain / cover / fill	contain	How the image fills the widget area. See fit mode descriptions below.
Background	Colour picker / blank	None	Colour shown behind the image in letterbox areas.

Image fit modes:

Mode	Description
contain	The full image is visible within the widget. Letterbox bars may appear if the image aspect ratio differs from the widget shape. Use for logos where clipping is unacceptable.
cover	The image fills the entire widget, cropping edges if needed to eliminate letterbox

	bars. Use for background textures or photographs where some cropping is acceptable.
fill	The image is stretched to fill the widget exactly, ignoring its original aspect ratio. Use with caution — distorts non-square images.

Tip: For sponsor logos, use contain with a transparent or matching background. For full-bleed background graphics, use cover. Keep logo files as PNG with a transparent background for the cleanest result.

Note: Large embedded images increase the layout JSON file size and may slow loading on first display. Compress images before uploading — a logo at 400 × 200 px is sufficient for most layouts.

7.4.3 Time of Day

Displays the current system time from the PolyField Track server. Uses a monospace font to prevent layout shift as digits change (so "1" takes the same width as "8"). Suitable for a permanent clock in the corner of a display layout.

Data source: PolyField Track server system clock. Updates every second.

Font: Monospace, to prevent layout shift as digits change.

Setting	Type	Default	Description
Format	HH:MM or HH:MM:SS	HH:MM:SS	Time format. HH:MM shows hours and minutes only; HH:MM:SS adds seconds.
Text colour	Colour picker	White	Time text colour.
Align	Left / Centre / Right	Centre	Horizontal alignment.
Background	Colour picker / blank	None	Widget background fill.

Tip: The Time of Day widget is most useful for scoreboards that operators leave up between events. It shows a live wall clock instead of a blank screen. Use HH:MM (no seconds) in layouts where the ticking seconds counter would be distracting.

7.5 Layout Tools

Layout tools are utility elements that affect how the layout behaves at run time but do not render visible content. They are only visible in the Layout Builder.

7.5.1 Overlay Area (Mask)

See Section 7.3.5 for full documentation. The Overlay Area appears in the Layout Tools group in the widget panel as well as in the Overlays group for ease of access.

7.6 Widget Colours Reference

Every widget supports a background colour setting. Additional colour settings vary by widget type.

Widget	Configurable colours	Notes
Results Table	Background	Row background, column header background, text colour
Results Table	Background	Configured per-row by theme; individual colour override available
Running Clock	Background, Running, Stopped, Armed, Time of Day	All four clock-state text colours are independently configurable
Clock Overlay	Background, Running, Stopped, Armed, Time of Day	Same as Running Clock
Event Name (Current)	Background, Text colour	Text colour applies to all clock states
Event Name (Result)	Background, Text colour	—
Wind (Current/Result)	Background, Text colour	—
Custom Text	Background, Text colour	—
Time of Day	Background, Text colour	—
Stopped Time	Background, Text colour	—
Logo / Image	Background	Background shows in letterbox areas around the image
Start List	Background	—
RAZA Results	Background	—
Overlays	Background (Clock Overlay only)	Text/Screensaver/Line View have no configurable colours

Tip: Set the background colour to match your layout's overall colour scheme so widget boundaries are invisible. Or set contrasting backgrounds on specific widgets (e.g. a dark background behind a Results Table) to create visual panels within the layout.

8 Web Views & Additional Displays

All web views are accessible from any device on the local network. No login required.

8.1 All Web URLs

URL (prefix with http://track.local:3000)	Purpose
/display	Main display screen (auto-assigns screen number on first load)
/remote	Remote panel for screen management from phone/tablet
/scoreboard	Full-screen scoreboard showing the scoreboard-assigned layout
/results	Multi-event results grid (all events in scrolling layout)
/athlete	Athlete search — find an athlete across all events
/speed	Speed dashboard — speedometer gauges and world record comparison
/raza	RAZA para-athletics combined ranking across all events
/layout-builder	Layout Builder design interface (best on desktop browser)

Tip: Replace track.local with the server IP address if mDNS is not available on the target device.

8.2 Multi-Result Mode (/results)

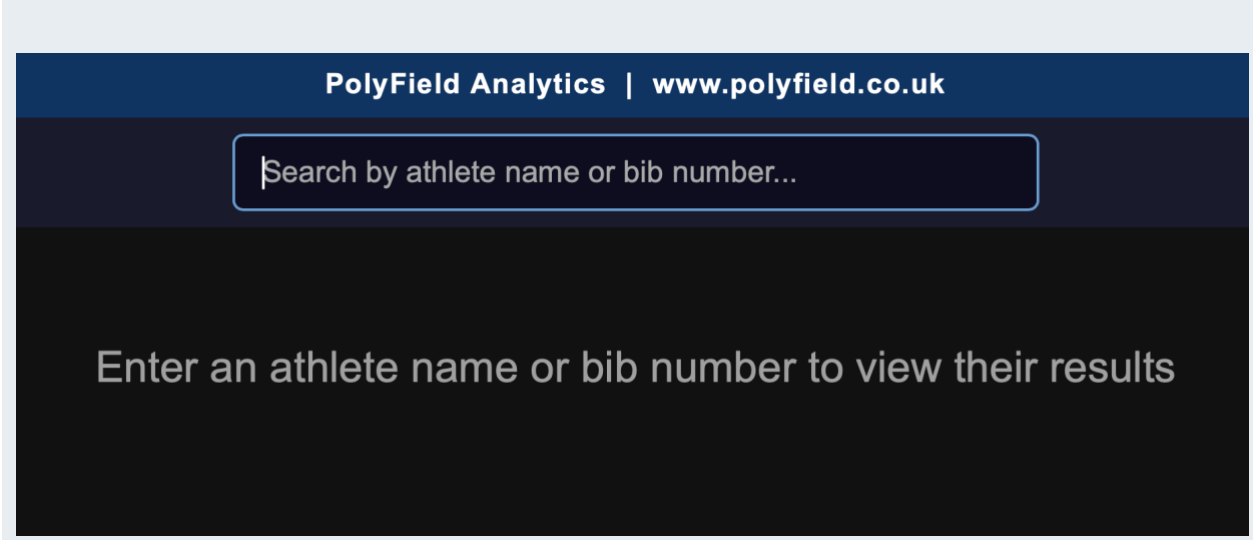
Displays all loaded events in a scrolling grid. Each event card shows the top results. Clicking an athlete name opens their profile in Athlete Search.

This view updates automatically every 3 seconds. Suitable for a secondary display showing all event history during a meeting.

8.3 Athlete Search (/athlete)

Allows spectators and coaches to look up an individual athlete's performance across all events at the meeting.

42. Type a name or bib number into the search box.
43. Results appear as a list of matching athletes.
44. Click an athlete to see all their events, places, times, and wind readings.
45. Click any event card to enlarge it to full screen.



This page is designed for public access on a separate display or via spectators' phones.

8.4 Speed Dashboard (/speed)

Displays speedometer-style gauges showing each athlete's calculated speed compared to the world record for that distance.

Features:

- Select any race from the dropdown to view its speed breakdown
- Each athlete gets a speedometer gauge scaled to the fastest time
- World record needles shown in blue (men's WR) and pink (women's WR)
- Percentage of world record shown for each athlete
- Speed units switchable between km/h, mph, and m/s

Speeds are calculated automatically from the distance in the event name and each athlete's finishing time.

> **Note:** Speed Dashboard is most effective for sprint events (60m – 400m). For longer events the gauge scale may compress.

8.5 RAZA Para-Athletics Rankings (/raza)

A full-meeting combined WPA RAZA ranking board. Shows all para-athletes who have competed, grouped by event distance and sorted by RAZA score.

How RAZA scoring works:

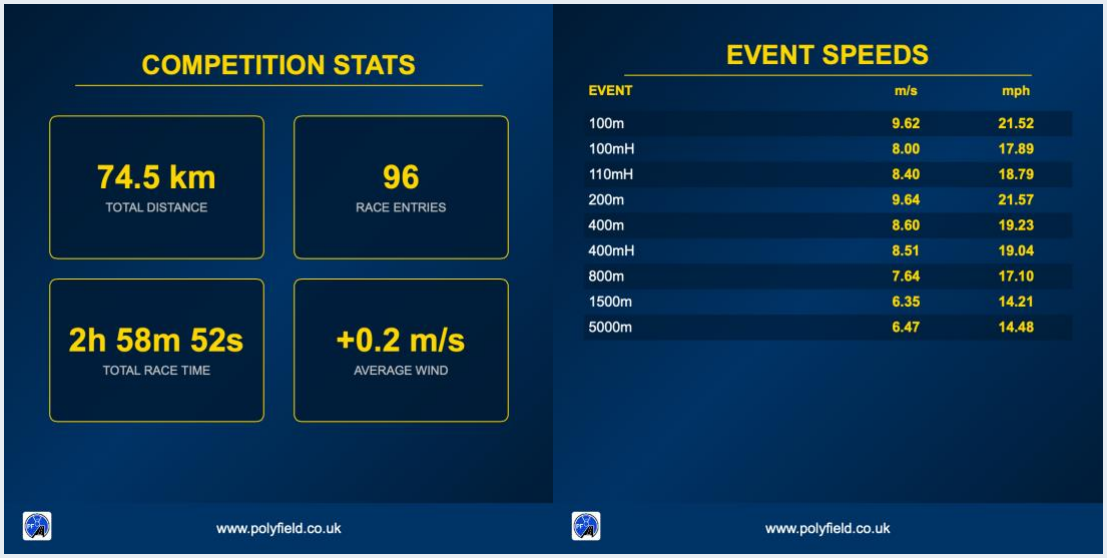
- Athletes are identified from lynx.evt using WPA class codes (e.g. WPA 34-M, WPA 54-W)
- Scores are calculated using the official 2025 WPA formula for each distance
- Only the best performance per athlete per distance is counted
- Rankings are combined across all WPA classes and updated live every 5 seconds

The footer of the /raza page shows the WPA formula used and a note about the classification method.

> **Tip:** For the /raza page to show results, athletes must have a WPA classification code in column 8 of lynx.evt and must have a valid finishing time in a loaded .lif file.

8.6 Social Graphics Generator

Generates branded competition statistics and event speed graphics, exportable as PNG for social media.



The generator has two views:

- **Competition statistics card (total distance, entries, race time, wind) at 1080 × 1080 px. Suitable for Instagram-style posts.:** Stats view
- **Event speed table comparing athletes' speeds by event, with pagination for many events:** Speeds view

To export:

46. Click Stats or Speeds to choose the graphic type.
47. Toggle distance units if needed (km / miles / metres).
48. Click Save PNG.

The PNG is saved to the results directory automatically. The filename is shown in a confirmation.

The PolyField logo and website URL are watermarked on each graphic.

9 Competition Statistics

The Competition Statistics section in the left panel aggregates data across all loaded results automatically.

Statistic	Unit	Description
Total Distance	metres	Sum of all event distances (parsed from event names like "100m", "400m Hurdles")
Race Entries	count	Total number of individual competitor results across all events
Total Race Time	H:MM:SS	Sum of all finishing times across all valid results
Average Wind	m/s	Mean of all wind readings from wind-reading events

Statistics update automatically whenever new results are loaded. They are also shown in the Social Graphics generator.

> **Note:** Distance parsing requires the event name to contain a recognisable distance (e.g. "100m", "200 Metres", "1500m"). Events without a recognisable distance are excluded from the total distance calculation.

10 Language Support

PolyField Track supports multiple languages. The language setting applies to both the desktop app interface and all web display pages simultaneously.

Language	Code	Notes
English	EN	Default language
French	FR	Français
Spanish	ES	Español

To change language: open the Language section in the left panel and click the desired language code. All connected display screens update within approximately 3 seconds.

Translated content includes: all UI labels and buttons, column headings (Place, Name, Time, Wind, etc.), clock states (READY, UNOFFICIAL TIME), layout builder terms, RAZA widget labels, and athlete/speed page text.

Note: If a translation key is missing for a given language, the English fallback is shown automatically.

Tip: This manual is designed to be translated. Table structure and screenshot captions are clearly labelled so translators can work from this Word document directly. See the separate translation workflow document if applicable.

11 Club Abbreviations

PolyField Track automatically shortens long club names in results tables to fit column widths. You can customise the abbreviations by editing club-list.csv.

11.1 How It Works

When a result is loaded, every club name is looked up in the club abbreviation table. If a match is found, the abbreviation is shown instead. If no match exists, the full name is shown (and may be truncated if very long).

11.2 Editing the Club List

49. In the desktop app, open the Settings section in the left panel.
50. Click Edit Club List.

This opens club-list.csv from the results directory in the system default application (typically Excel or a text editor).

Format: one entry per line, full club name and abbreviation separated by a comma:

```
Kingston Athletics Club,KAC
South London Harriers,SLH
Thames Valley Harriers,TVH
```

51. Save the file after editing.

Changes take effect immediately on the next display refresh (within 10 seconds). No app restart required.

> **Note:** The file is created automatically with an empty header row if it does not exist. The header row (Club Name,Abbreviation) is ignored by the parser.

> **Tip:** Club names are matched case-insensitively. "Kingston Athletics Club" and "kingston athletics club" both match the same entry.

13 Data File Reference

This chapter provides a technical reference for the result and event file formats that PolyField Track reads. It is intended for operators configuring their timing system export, or troubleshooting cases where data is not displaying as expected.

13.1 FinishLynx .lif Format

The .lif (Live Information File) format is the primary result file produced by FinishLynx. It is a comma-separated (CSV) file using Windows-1252 or UTF-8 character encoding.

File Structure

Row 0 — Event header:

Field	Content
0	Record type (typically "0")
1	FinishLynx event reference number
2	Round and heat (e.g. "F-1")
3	Event name (e.g. "Men's 100m Final")
4	Wind reading in m/s (e.g. "+1.8" or "(Manual) 0.4"). "Manual" and parentheses are stripped automatically.

Rows 1+ — Competitor rows (one per finisher):

Field	Content
0	Place number (1, 2, 3 ...). "DQ" or "DNF" for disqualified/did not finish. Empty or "DNS" rows are skipped.
1	Bib / ID number
2	Lane number
3	Last name
4	First name
5	Club / affiliation
6	Finishing time (e.g. "10.52", "1:52.34", "2:03:14.00"). Times are rounded up to the next hundredth of a second for display.

Note: PolyField Track accepts .lif files in Windows-1252, ISO-8859-1, UTF-8, UTF-16LE, and UTF-16BE encodings. Encoding is detected automatically.

Tip: FinishLynx writes the .lif file each time a result is confirmed. If a race is re-scored, the file is overwritten and PolyField Track displays the updated result automatically.

Example .lif File

```
0,1,F-1,Men's 100m Final,+1.8
1,101,3,Smith,James,Kingston AC,10.52
2,207,5,Jones,David,Thames VH,10.61
3,315,7,Brown,Mark,South London,10.74
,042,2,Williams,Chris,Belgrave H,DQ
```

13.2 FinishLynx lynx.evt Format

The lynx.evt file is the FinishLynx schedule and athlete database. PolyField Track reads it to populate the Start List widget and to identify WPA-classified para-athletes for RAZA scoring.

lynx.evt is a comma-separated file in Windows-1252 encoding. It contains two types of rows that alternate:

Event Rows (first field non-empty)

Field	Content
0	Event reference number (e.g. "1", "2")
1	Round number (e.g. "1")
2	Heat number (e.g. "1")
3	Event name — full description (e.g. "Men's 100m Final 1630h")
4+	Scheduled time and other metadata (not used by PolyField Track)

Athlete Rows (first field empty)

Field	Content
0	Empty (distinguishes athlete rows from event rows)
1	Bib / ID number
2	Lane number (empty if not assigned)
3	Last name
4	First name
5	Club / affiliation
6	Scratch/entry status (not used by PolyField Track)
7	Seed mark (not used by PolyField Track)
8	WPA classification code, e.g. "WPA 34-M". Empty for non-para athletes. See Section 13.5.

Event Name Matching

When FinishLynx broadcasts an event name via LSS, PolyField Track normalises both the UDP name and the lynx.evt event names before comparing them:

- Trailing scheduled time tags (e.g. " 1649h") are stripped
- Version suffixes (e.g. " (version 1)") are stripped
- The comparison is case-insensitive

This means "U17 M-200m_f-1 1649h (version 1)" in lynx.evt will correctly match "U17 M-200m_f-1" from the UDP feed.

Example lynx.evt Rows

```
1,1,1,Men's 100m Final  1630h,16:30
,101,3,Smith,James,Kingston AC,,10.52,
,207,5,Jones,David,Thames VH,,10.61,WPA 34-M
2,1,1,Women's 100m Final  1645h,16:45
,301,4,Williams,Sarah,Belgrave H,,11.20,WPA 54-W
```

13.3 Timetronics .res Format

The .res format is the standard Timetronics results export. It is a TAB-delimited file, typically in Windows-1252 encoding.

File Structure

Row 0 — Image information (TAB-delimited):

Field	Content
0	Filename of the corresponding photo finish image
1	Wind reading (e.g. "+1.2"). The string "m/s" is appended automatically if not already present.
2	File size (not used by PolyField Track)
3	Lines per second (not used by PolyField Track)
4+	Timestamp and date metadata (not used by PolyField Track)

Optional event name override — any row beginning with # Event: sets the event name:

```
# Event:	Men's 100m Final
```

If no # Event: row is present, the event name is derived from the filename (extension stripped).

Row 1 — Column headers (skipped by PolyField Track)

Rows 2+ — Competitor rows (TAB-delimited):

Field	Content
0	Place number (1, 2, 3 ...). "DQ" or "DNF" for special results. Empty or "DNS" rows are skipped.
1	Lane number (not used in the display)
2	Finishing time (e.g. "10.52", "1:52.34"). Rounded up to next hundredth of a second.
3	Athlete ID / bib number
4	Athlete name (full name as a single field, e.g. "James Smith")
5	Club / affiliation

Name Handling

Timetronics .res files store the athlete name as a single field (First Last). PolyField Track splits this at the first space to derive first and last name. If the name has no space, it is treated as the last name only.

Example .res File

```
100m-Men-Final.jpg	+1.8		16:30:00	2024-07-06
# Event:	Men's 100m Final
Place	Lane	Time	ID	Name	Club
1 3	10.52 101	James Smith Kingston AC
2 5	10.61 207	David Jones Thames VH
DQ	2	DQ	42	Chris Williams	Belgrave H
```

13.4 Timetronics .txt Format

The .txt format is a variant of the Timetronics export used by some configurations. It has the same TAB-delimited structure as .res with the following differences:

Difference from .res	How PolyField Track handles it
Wind field (row 0, field 1)	May contain "N/A" or "N/A m/s". These values are stripped and the wind reading is left blank.
ID field (row 2+, field 3)	May contain bracketed content such as "(club code)" which is automatically stripped. The remaining text is the athlete ID.
Club field (row 2+, field 5)	Field 5 in .txt files is an "Information" field containing numbers, not a club name. PolyField Track automatically ignores this field — no club/affiliation is shown for .txt results.

Note: Because .txt files do not provide club data, the Club column in the Results Table widget will be blank for .txt results. If club display is important, use .res export format instead of .txt.

13.5 WPA Classification Codes (lynx.evt)

WPA classification codes in column 8 of lynx.evt identify para-athletes and their functional classification for RAZA scoring. PolyField Track recognises codes in the format:

WPA · [class number] · - · [gender letter]

For example: WPA 34-M, WPA 54-W, WPA 53-V

lynx.evt value	Class number	Gender code	Interpreted as
WPA 34-M	34	M (Men)	T34 class, male athlete
WPA 54-W	54	W (Women)	T54 class, female athlete
WPA 53-V	53	V (Mixed)	T53 class, gender-neutral or mixed
WPA 46-M	46	M (Men)	T46 class, male athlete
WPA 12-W	12	W (Women)	T12 class, female athlete

The gender letter determines how RAZA scores are computed:

- **Men's RAZA parameters applied:** M
- **Women's RAZA parameters applied:** W
- **Treated as mixed/unspecified; both men's and women's parameters are tried and the best match is used:** V or any other letter

Class numbers map to WPA T-class designations:

T11–T13, T20, T31–T38, T40–T47, T51–T54, T61–T64. Classes outside the supported range (no RAZA parameters available) are excluded from scoring.

Note: Athletes must have a finishing time in a loaded .lif result file AND a WPA code in lynx.evt to appear in RAZA Results. Athletes in .res/.txt result files are not scored because RAZA requires the EVT file to identify WPA classification.

Tip: Enter WPA codes in FinishLynx by editing the athlete record in the schedule. The code goes in the "Affiliation 2" or custom field that exports to column 8 of lynx.evt. Check with your FinishLynx documentation for your version's exact field mapping.

14 Day-of-Competition Checklist

Use this checklist to ensure PolyField Track is fully set up before competition begins.

Before Arrival

- ☐ Copy any screensaver/logo images to a USB drive or the results directory

On Arrival

- ☐ Launch PolyField Track
- ☐ Click Select Results Folder and choose the FinishLynx results directory
- ☐ Confirm "Monitoring directory for new results" appears in the app
- ☐ Configure FinishLynx LSS scoreboard output to port 5001
- ☐ Run a test event in FinishLynx — confirm the Running Clock appears in PolyField Track
- ☐ Connect display screens by opening the display URL on each device
- ☐ Verify each screen appears in the Screens panel as Live
- ☐ Assign layouts to screens from the Screens panel or Remote panel
- ☐ Set language if not English
- ☐ Set rotation mode appropriate for event format (Page recommended for field events)
- ☐ Link a screensaver image if using between events
- ☐ Test text overlay: type a message and click Display, then Clear
- ☐ Confirm KACPH Scoreboard preview is showing correctly

During Competition

- ☐ Results appear automatically when FinishLynx writes .lif files — no action required
- ☐ Use Clock Show / Clock Hide for clock overlay as needed
- ☐ Use Line View after close finishes while reviewing photo finish
- ☐ Use Display Text for announcements between events
- ☐ Use Screensaver to show sponsor logos / holding graphics between events
- ☐ Reassign layouts from Remote panel if screen needs changing mid-session

After Competition

- ☐ Export custom layouts as backup if edited during the day
- ☐ Save social graphics PNG if required
- ☐ Close display screens on all devices

15 Troubleshooting

Display screen shows "Could not connect to server"

- Check that PolyField Track is running on the timing computer
- Confirm both devices are on the same Wi-Fi or wired network
- Try using the IP address instead of track.local (IP shown in Screens panel)
- Check the timing computer firewall allows inbound connections on port 3000

Results not updating on display screens

- Check the results directory is correctly linked (app header should say "Monitoring directory")
- Confirm FinishLynx is writing .lif files to the same directory (or Timetronics is writing .res/.txt files)
- Check the Debug Log (toggle in left panel) for file detection messages
- Try clicking Restore Last Result to manually reload
- Ensure the results file is complete before checking — PolyField Track loads files as soon as they are written; a partially written file may be rejected

Timetronics results not appearing

- Confirm Timetronics is configured to write .res or .txt files to the same directory selected in PolyField Track
- Check the file extension is exactly .res or .txt (case-insensitive)
- Verify the results file has at least 3 rows (info row, header row, one competitor row)
- Check the Debug Log for any file parsing errors
- If club names are blank, this is expected for .txt files — see Section 13.4
- Running Clock, Start List, and RAZA scoring are not available with Timetronics — see Section 2.4

Running Clock not appearing

- Confirm FinishLynx LSS scoreboard is configured to output on port 5001
- Check that the LSS scoreboard is enabled in FinishLynx Options → Scoreboard
- Confirm the timing computer firewall allows inbound UDP on port 5001
- The clock widget shows Time of Day when no UDP has been received — if the time is correct, the widget itself is working

track.local not resolving on Windows display devices

mDNS / Bonjour may not be enabled by default on Windows. Solutions:

- Use the server IP address directly instead of track.local
- Install Bonjour for Windows (available from Apple)
- Enable mDNS via PowerShell: Set-Service "Bonjour Service" -StartupType Automatic; Start-Service "Bonjour Service"

Screen shows as Stale

A screen is Stale if it has not sent a heartbeat in the last 15 seconds.

- Refresh the browser on the display device
- Check the display device's network connection
- If the page is visible but stale, the screen may be in a background tab — bring it to the foreground

Layout not appearing on a screen

- Confirm the layout has been assigned to the screen in the Screens panel (not just activated globally)
- Check the layout has been saved (status shows "Saved")
- Refresh the display screen browser

RAZA results not appearing

- Confirm lynx.evt is in the results directory and is readable
- Check athlete records in lynx.evt have a WPA class code in column 8 (e.g. WPA 34-M)
- Confirm the current .lif file contains those athletes' bib numbers
- Check the Debug Log for EVT parsing messages

Club names showing truncated instead of abbreviations

- Check club-list.csv exists in the results directory
- Confirm the club name in the CSV matches exactly (case-insensitive) what FinishLynx writes to the .lif file
- Try clicking Edit Club List and saving the file again to trigger a reload

Photo finish images not showing in Line View

- Confirm FinishLynx is configured to save JPG images to the results directory
- Check the results directory contains .jpg or .jpeg files
- The Line View button is disabled if no JPGs are found — check the directory path

16 Quick Reference Card

Key controls and URLs for day-of-competition use.

Key Controls

Control	Action
Select Results Folder	Link FinishLynx results directory
Restore Last Result	Reload previously displayed result
Display (text button)	Send typed text to all screens
Clear	Remove text or screensaver from all screens
Screensaver	Push linked image to all screens
Clock Show / Clock Hide	Toggle full-screen clock overlay
Line View	Toggle photo finish / results alternation
Layout Builder	Open layout design interface
Open Remote	Open /remote in browser for tablet management
Full Screen App	Maximise entire app window (Esc to exit)
Full Screen Table	Maximise results table only (Esc to exit)

Keyboard Shortcuts

Key	Action
Esc	Exit full-screen mode

Web URLs

URL	Purpose
http://track.local:3000/display	Display screen (browser, any device)
http://track.local:3000/remote	Remote management panel
http://track.local:3000/scoreboard	KACPH Scoreboard view
http://track.local:3000/results	All events grid
http://track.local:3000/athlete	Athlete search
http://track.local:3000/speed	Speed dashboard
http://track.local:3000/raza	RAZA para-athletics rankings
http://track.local:3000/layout-builder	Layout Builder (desktop browser)

FinishLynx LSS Configuration

Setting	Value
Type	Network
Protocol	LSS
IP	Broadcast address (e.g. 192.168.0.255) or 127.0.0.1
Port	5001